

The Murty-Simon bound at order thirty

Reviewer edition 3 - hand lemmas and explicit integer tables

Paul Lenz research project - developed with ChatGPT/Geeps

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The Murty-Simon bound at order thirty

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Abstract

We present a candidate proof that every finite simple diameter-two edge-critical graph on thirty vertices has at most 225 edges, with equality exactly for the complete bipartite graph $K_{15,15}$. The proof uses a complement construction with selected and residual cross-edges, a threshold-capacity inequality, hand clipping lemmas, and explicit finite integer tables. A source-independent twelve-label bound closes every dense case of maximum degree at least seventeen. The degree-fifteen equality case has an elementary proof. At maximum degree sixteen, a hand classification of 100 demand profiles leaves eight source inequalities at 226 edges and 211 envelope evaluations at 225 edges. Every required mathematical argument and finite table is included in this manuscript. The verification companion records reproducibility, provenance and the boundary of internal checking.

Status. Complete candidate mathematics; exact internal arithmetic REPRODUCED; independent specialist mathematical review OPEN. Separately written checks by the same assistant are internal corroboration, not external validation. This edition does not change the governed theorem ledger, assert novelty, or claim the unrestricted conjecture.

Theorem 1 (candidate). Let G be a finite simple graph on thirty vertices with diameter two such that deleting any edge increases its diameter. Then

$$e(G) \leq 225, \quad e(G) = 225 \iff G \cong K_{15,15}.$$

Distances between disconnected vertices are infinite. Write $m = e(G)$ and $\Delta = \Delta(G)$. A histogram such as $(1^7, 2, 3^8)$ means seven entries equal to one, one equal to two, and eight equal to three. Empty multiplicities are omitted. Indicator functions have values zero or one.

How to read the proof

Sections 1-8 give the complete assembly. Appendices A-H supply its arguments and finite arithmetic without requiring another paper or a computational acceptance result. The appendix order follows the dependencies:

Appendix	Material	Role in the proof
A	Selected/residual bridge	Universal graph-theoretic implications
B	Twelve-label bound	Hand closure of all dense degrees at least 17
C	Thirteen-label classification	Hand reduction to exactly 100 profiles
D	All five- and six-lift intervals	Proof-critical finite classification arithmetic
E	Residual-tail reconstruction	Completeness of the endpoint row lists
F	Eight source inequalities	Proof-critical arithmetic at 226 edges
G	Four monotone envelopes	Formulas and local domains at 225 edges
H	All 211 positive-gap rows	Proof-critical endpoint arithmetic

The larger profile sweeps and earlier LP/Farkas systems are not premises. **The printed finite tables are premises to be checked.** All such tables occur here, including rejected preimage intervals and every endpoint gap. A successful checker run cannot replace review of the universal bridge.

Notation is scoped locally. In the tail lemmas $p = N_1$ counts positive demands; in graph formulas p_u is supplement indegree. Section 5 and Appendix E use λ for ledger slack. Appendix G and its envelope evaluations use λ for a certificate coefficient after ledger slack has been set to zero. Bridge equation numbers and section references are local to Appendix A; analogous conventions apply to the other appendices.

1. Degree entry and the equality case Delta=15

If $m \geq 226$, degree sum gives $\Delta \geq 16$. If $m = 225$, it gives $\Delta \geq 15$.

Suppose $m = 225$ and $\Delta = 15$. Every vertex then has degree fifteen. For any nonadjacent pair x, y , their two neighborhoods lie among the other twenty-eight vertices and have total size thirty, so they have at least two common neighbors.

Deleting a single edge can destroy at most one common neighbor of a fixed nonadjacent pair. Thus every originally nonadjacent pair remains at distance at most two after any one edge deletion. Every originally adjacent pair except the deleted edge's endpoints remains adjacent. Consequently, for an edge uv to be critical, its endpoints must have no common neighbor. Otherwise even that pair remains at distance two after deletion, contradicting criticality.

Every edge therefore lies in no triangle, so G is triangle-free. Choose an edge uv . The sets $N(u)$ and $N(v)$ are disjoint independent sets, each of size fifteen, and together contain every vertex. Every vertex has degree fifteen and can have neighbors only in the opposite set of size fifteen. All cross edges are therefore present, giving $G = K(15, 15)$.

Conversely, $K(15, 15)$ has diameter two, and deleting a cross edge makes its endpoints distance three. This proves the entire $\Delta = 15$ equality branch directly. **No published dominating-edge theorem is a dependency of this assembly.** The same elementary argument applies to a d -regular diameter-two edge-critical graph on $2d$ vertices for $d \geq 2$.

2. The universal bridge used here

Choose a minimum-degree vertex v in the complement H and put

$$A = N_H(v), \quad a = 29 - \Delta, \quad B = V(H) \setminus N_H[v], \quad b = \Delta.$$

The canonical parameterized bridge (Appendix A) supplies one selected cross-edge per missing unordered B -pair, with every other cross-edge residual. For labels use F -degree d_i , residual degree R_i , selected degree x_i and demand $s_i = \max(0, d_i - R_i)$. For sources use residual degree ρ_u , selected outdegree q_u and supplement indegree p_u . With $t = m - b(a+1)$, its needed conclusions are:

$$e(F) = r + t, \quad r = \sum R_i = \sum \rho_u, \quad S = \sum s_i \geq r + 2t, \quad x_i \geq s_i.$$

For $t > 0$, every source has $\rho_u \geq 1$. Every selected incidence ui has

$$s_i \leq \rho_u, \quad d_i \leq \rho_u + q_u - 1, \quad R_i + x_i \geq q_u + p_u.$$

Every selected source-to-supplement arc $u \rightarrow w$ has $\rho_w + q_w \geq q_u - 1$. Source degree and simple-graph capacity give

$$q_u \leq a - \rho_u, \quad p_u \leq \rho_u + b - a - 1, \quad q_u + p_u \leq b - 1.$$

Threshold capacity gives $W_h \leq C_h(z_h)$, where $z_h = \#\{\rho \geq h\}$, $W_h = \sum_{s_i \geq h} s_i$, and $C_h(z) = \lfloor z(z-1) + h(h+1) \rfloor / 2$.

The internal bridge audit, reproduced in the verification companion, checks the selected-edge injection, forced residual edges, residual activity, threshold pair counting, and the precise source domain. Charging and the isolated- C improvement are **not** needed in this assembly. Appendix A includes the corrected source-degree identity and the precise unique-exception argument.

3. $\Delta \geq 17$ has a hand exclusion

For $\Delta = 17, \dots, 28$, $1 \leq a \leq 12$, $b \geq 17$, and at every $m \geq 225$ the surplus $t = m - b(30 - b)$ is positive.

The source-independent twelve-label lemma (Appendix B), applied after padding the demand vector with zeros, gives $Q \leq 18$. The bridge gives $Q \geq b + 2t \geq 19$, a contradiction. More sharply, at $\Delta = 17$ it gives $m \leq 221$.

If $\Delta = 29$, a universal vertex exists. Any edge between two other vertices could be deleted while the universal vertex still kept all pairs within distance two. Criticality therefore makes G a star, with only twenty-nine edges.

Thus every degree at least seventeen is excluded at the dense scopes. No historical $\Delta = 17$ profile checks, final Hall duals, or higher-degree charging scans remain necessary here.

4. $\Delta = 16$: the complete profile input is hand classified

Here $(a, b) = (13, 16)$ and $t = m - 224$. In the dense scopes t is positive. The hand classification (Appendix C) proves:

- $Q \leq 21$ for every threshold-admissible demand vector;
- exactly 100 profiles have $Q \geq 18$;
- the score distribution is 64 at 18, 29 at 19, 6 at 20, 1 at 21;
- the unique $Q = 21$ vector is (3^{13}) .

Its proof clips demands to four, classifies seventy profiles, recovers thirty five-containing lifts, and rules out all higher preimages. Every accepted and rejected preimage interval is printed. The historical exhaustive list is only corroboration.

Since the bridge requires $Q \geq 16 + 2t$, every $t \geq 3$, equivalently $m \geq 227$, is impossible. Only $t = 2$ and $t = 1$ remain.

5. Residual rows from at most three slack units

For $h = 2, \dots, 13$, let z_h be the residual-degree tails and put g_h equal to the minimum threshold for the demand tail; $g_{13} = 0$. Residual activity gives $r = 16 + \sum z_h$. The exact slack identity is

$$Q - (16 + 2t) = \lambda + \sum_{h=2}^{13} (z_h - g_h), \quad \lambda = S - r - 2t \geq 0.$$

All summands are nonnegative integers, and $16 \geq z_2 \geq \dots \geq z_{13} \geq 0$. Such tails determine ρ uniquely. This is the written threshold-slack reconstruction (Appendix E), with the new hand classification supplying completeness.

At $t = 2$, the seven profiles with $Q \geq 20$ give nine rows. Six $Q = 20$ profiles have no slack. For (3^{13}) , $g_2 = g_3 = 9$ and all higher g are zero; its one unit of slack can be assigned to λ , to z_2 , or to a new $z_4 = 1$. There are no other monotone possibilities.

At $t = 1$, at most three units are distributed, giving 272 rows. The independent audit reconstructs both sets by adding boxes to monotone tail diagrams, using no historical residual-row generator. It agrees exactly with the printed prior frontier.

Whenever every demand is positive, $s_i = d_i - R_i$ pointwise, and the ledger gives $S = r + 2t$. Hence $\lambda = 0$. This immediately excludes the one positive-slack $t = 2$ row and all 61 positive-slack $t = 1$ rows. It leaves eight rows at $m = 226$ and 211 at $m = 225$. This implication uses positivity, not an assumed demand equality; it resolves an imprecise sentence in the earlier $m = 226$ exposition.

6. The eight m226 rows

The hand endpoint note (Appendix F) states all eight rows and their small source certificates. Here is its algebraic mechanism.

For a tight row, put $e_i = x_i - s_i \geq 0$, $E = \sum e_i$, $M = \sum x_i = \sum q_u$, and $\alpha_u = \max(0, p_u - \rho_u + 1)$. The selected incidence inequalities imply $e_i \geq \alpha_u$; the valid source bound $p_u \leq \rho_u + 2$ implies $\alpha_u \leq 3$. Thus, with $c = \max s_i + 3$,

$$B_0 := \sum q_u \alpha_u \leq \sum_i (s_i + e_i) \min(e_i, 3) \leq cE.$$

For each k , supplement forcing gives

$$H_k = \sum_{q_u \geq k+1} q_u - \sum_{\rho_w + q_w \geq k} p_w \leq 0.$$

The eight explicit source inequalities in the cited table give $B_0 - cE \geq \delta$, with respective positive deltas

$$1, 3, 12, 3, 4, 6, 1, 3/2.$$

Their direct integer minima use $0 \leq q \leq 13 - \rho$, $0 \leq p \leq \rho + 2$, and $q = 0$ when ρ is below every demand. The independent implementation verifies all 859 source states in that table. Each row contradicts $B_0 \leq cE$, completing m226.

7. The 211 m225 rows

For every tight row, including the four with a zero demand, equality in $\sum \max(0, d_i - R_i) \geq \sum (d_i - R_i)$ forces $d_i = R_i + s_i$ at every label.

The four-envelope argument (Appendix G) uses the exact local domains

$$0 \leq R \leq 12 - s, \quad s \leq x \leq \min(16 - R, \#\{u : \rho_u \geq s\}),$$

$$0 \leq q \leq \min(13 - \rho, \#\{i : s_i \leq \rho\}), \quad 0 \leq p \leq \rho + 2.$$

These are necessary simple-graph capacities. The stronger historical isolated-C degree bound is not used.

On a selected incidence the label coordinates $(s, R+s, 16-R-x)$ are componentwise at most the source coordinates $(\rho, \rho+q-1, 16-q-p)$. A nondecreasing potential therefore gives

$$\sum_i x_i \Phi(s_i, R_i + s_i, 16 - R_i - x_i) \leq \sum_u q_u \Phi(\rho_u, \rho_u + q_u - 1, 16 - q_u - p_u).$$

Combine this with $\sum R = r$, $\sum x = \sum q = \sum p$ and the supplement tails $H_k \leq 0$. For the four printed nonnegative integer coefficient sets, take local minima ℓ_s and σ_ρ of the resulting label and source expressions. Any actual graph must satisfy

$$\sum_s n_s \ell_s + \sum_\rho n_\rho \sigma_\rho - \lambda r \leq 0.$$

Here λ denotes the certificate's residual-budget coefficient, distinct from the already-zero ledger slack in Section 5. The complete 211-row arithmetic appendix (Appendix H) displays a strictly positive left side for every row. Certificates B1, B2, B3, C1 cover $195+13+2+1=211$ rows; the minimum assigned integer gap is one.

The new independent audit recomputes all 12,772 local states with a separately written potential evaluator and reconstruction. It matches all 844 original per-template row gaps exactly. Every minimum can also be checked from the printed endpoint formulas in the four-envelope note. This is explicit finite integer arithmetic, not a numerical infeasibility conclusion.

Thus no $\Delta=16$ graph has m_{225} or m_{226} ; Section 4 excluded all larger edge counts.

8. Complete assembly and remaining review boundary

At $m \geq 226$, $\Delta \leq 15$ is excluded by degree sum; $\Delta=16$ by Sections 4-7; $\Delta=17, \dots, 28$ by Section 3; and $\Delta=29$ by the star argument. Hence $m \leq 225$.

At m_{225} , $\Delta \leq 14$ is excluded by degree sum, $\Delta=15$ forces $K(15, 15)$ by Section 1, and all larger degrees are excluded above. The converse was verified in Section 1. This completes the candidate proof of Theorem 1.

The logical route now uses neither Fan's density theorem nor the published dominating-edge reduction, and it retains no $\Delta=17$ or higher-degree computational branch. The remaining substantial finite audit surface is the explicit $\Delta=16$ classification, tail reconstruction and endpoint arithmetic.

Appendix A. Universal selected/residual bridge

A.1. Setup

Let G be a finite simple diameter-two edge-critical graph on n vertices. Put

$H = \text{complement}(G)$.

Choose a minimum-degree vertex v of H and write

$A = N_H(v)$,
 $a = |A| = \delta(H) = n-1-\Delta(G)$,
 $B = V(H) \setminus N_H[v]$,
 $b = |B| = \Delta(G)$.

Thus

$n=a+b+1$.

Let

$C=H[A]$,
 $F=\text{complement}(C)$ on A .

For i in A , let $d_i=d_F(i)$.

Write $m=e(G)$ and define the surplus over the complete bipartite benchmark determined by B as

$t = m-b(n-b) = m-b(a+1)$.

The strongest conclusions below assume $t>0$. The quasi-edge construction and exact ledger do not.

A.2. Complement form of diameter-two edge-criticality

Take an edge uw of G , equivalently a missing pair uw of H . Deleting uw from G is the same as adding uw to H .

Because G is diameter-two edge-critical, in $G-uw$ some pair x, y has distance greater than two. Equivalently, in $H+uw$, the pair x, y is adjacent and its open neighbourhoods together contain every vertex: it is an adjacent total-dominating pair.

No adjacent total-dominating pair existed already in H , because that would correspond to a pair at distance greater than two in G . Hence the newly total-dominating pair must use u or w , the only vertices whose H -neighbourhoods changed.

Now suppose u, w in B . The new total-dominating pair cannot be $\{u, w\}$, because both miss v . Hence, after interchanging u and w if necessary, there is an existing cross-edge ui with i in A such that

$$N_H(u) \cup N_H(i) = V(H) \setminus \{w\}. \quad (2.1)$$

Write

$ui \rightarrow w$.

The vertex w is the unique exception of the adjacent pair $\{u, i\}$ in H .

A.3. One selected representative per missing unordered B-pair

For every missing **unordered** pair $\{u, w\}$ of $H[B]$, choose exactly one cross-edge supplied by Section A.2, after orienting the pair if necessary.

Call these chosen A-B edges **selected**. Call every other existing A-B edge **residual**.

A selected edge $ui \rightarrow w$ determines:

- its B-source u ;
- its A-label i ;

- its unique B-exception w .

Therefore the selected cross-edge recovers the missing unordered B-pair $\{u, w\}$. Consequently:

1. distinct missing unordered B-pairs cannot choose the same selected cross-edge;
2. the two orientations of one missing B-pair cannot both be selected;
3. at a fixed source u , selected labels are distinct and their supplements/exceptions are distinct.

For u in B define

ρ_u = residual degree of u into A ,
 q_u = number of selected edges with source u ,
 p_u = number of selected missing B-pairs whose exception is u .

For i in A define

R_i = residual degree of i into B ,
 x_i = selected degree of i into B .

Let

$$r = \sum_{u \in B} \rho_u = \sum_{i \in A} R_i.$$

A.4. Exact ledger

Every unordered pair of B is exactly one of:

- an edge of $H[B]$; or
- a missing pair of $H[B]$, carrying exactly one selected representative.

Hence

$$\# \text{selected cross-edges} + e(H[B]) = C(b, 2). \quad (4.1)$$

Also

$$\begin{aligned} e(H) &= C(n, 2) - m \\ &= a + e(C) + r + \# \text{selected} + e(H[B]). \end{aligned}$$

Using (4.1), $n=a+b+1$, and $t=m-b(a+1)$, this rearranges exactly to

$$e(C) + r = C(a, 2) - t. \quad (4.2)$$

Since F is the complement of C on A ,

$$e(F) = r + t, \quad (4.3)$$

$$\sum_i d_i = 2(r + t). \quad (4.4)$$

No inequality has been used here.

A.5. Label demand

Because v has minimum H -degree a , every i in A satisfies $d_H(i) \geq a$. Its H -neighbours consist of:

- v ;
- $a - 1 - d_i$ neighbours inside A ;
- $R_i + x_i$ neighbours in B .

Thus

$$1 + (a - 1 - d_i) + R_i + x_i \geq a,$$

so

$$x_i \geq d_i - R_i.$$

Define

$$\begin{aligned} s_i &= \max(0, d_i - R_i), \\ S &= \sum_i s_i. \end{aligned}$$

Then

$$x_i \geq s_i. \quad (5.1)$$

Summing the weaker inequalities $s_i \geq d_i - R_i$ and using (4.4) gives

$$S \geq 2(r+t) - r = r + 2t. \quad (5.2)$$

A positive demand s_i therefore corresponds to at least s_i distinct selected A-B incidences, hence s_i distinct B-sources.

A.6. Pointwise forcing from a selected edge

Fix a selected edge

$$u_i \rightarrow w.$$

A.6.1. Label residual forcing

Let j be an F-neighbour of i , so ij is absent from $H[A]$. Since j is an A-vertex and w is the unique exception in (2.1), uj must be an H-edge.

If uj is residual, charge j to a residual source incidence at u .

Otherwise uj is selected, say $uj \rightarrow z$. The supplement z differs from w because the missing unordered pair $\{u, w\}$ already has its unique selected orientation. Since $ui \rightarrow w$ must dominate z and uz is missing in $H[B]$, iz is an H-edge.

Moreover iz is residual. Indeed, ij is absent, and because z is the exception of $uj \rightarrow z$, jz is absent. Thus both endpoints of iz miss the A-vertex j . A selected A-B edge has its unique exception in B and must dominate every A-vertex, so iz cannot be selected.

Distinct selected j at u have distinct supplements z . Therefore the F-neighbours of i inject into at most ρ_u residual source incidences and R_i residual label incidences:

$$d_i \leq \rho_u + R_i. \quad (6.1)$$

In particular, at every selected incidence of label i ,

$$s_i \leq \rho_u. \quad (6.2)$$

A.6.2. Supplement residual forcing

For the same F-neighbour j , again uj is an H-edge. If it is residual, charge it at u . Otherwise write $uj \rightarrow z$.

The supplements z and w are distinct. The selected pair $\{u, j\}$ must therefore dominate w . Since uw is missing, jw is an H-edge.

This edge is residual: ij is absent and iw is absent because w is the exception of $ui \rightarrow w$, so both endpoints j and w miss the A-vertex i . Distinct j give distinct residual edges at w . Hence

$$d_i \leq \rho_u + \rho_w. \quad (6.3)$$

A.6.3. Source selected-degree forcing

Every F-neighbour j of i is a cross-neighbour of u . Source u has exactly $\rho_u + q_u$ A-neighbours, one of which is i . Therefore

$$d_i \leq \rho_u + q_u - 1. \quad (6.4)$$

A.6.4. Supplement forcing

For every other selected edge $uj \rightarrow z$ at u , its exception z differs from w . Thus $\{u, j\}$ must dominate w . Since uw is missing, jw is an H-edge.

The $q_u - 1$ other selected labels j are distinct. They therefore give $q_u - 1$ distinct A-neighbours of w , each either residual or selected from source w . Hence

$$\rho_u + q_u \geq q_u - 1. \quad (6.5)$$

A.7. Exact missing degree in B and source capacities

Every missing unordered B-pair incident with u is selected in exactly one orientation: outward from u or inward to u . Therefore

$$q_u + p_u = \text{missing degree of } u \text{ in } H[B]. \quad (7.1)$$

The H-degree of u is

$$\begin{aligned} d_H(u) &= (\rho_u + q_u) + [(b-1) - (q_u + p_u)] \\ &= \rho_u + b - 1 - p_u. \end{aligned}$$

Minimum H-degree gives

$$p_u \leq \rho_u + b - a - 1. \quad (7.2)$$

Also

$$q_u + p_u \leq b - 1, \quad (7.3)$$

$$q_u + \rho_u \leq a. \quad (7.4)$$

Equation (7.4) is simply the A-side capacity at source u .

A.8. Endpoint load

Again fix $u_i \rightarrow w$.

The following are distinct B-neighbours of label i :

1. u itself;
2. the $q_u - 1$ supplements of the other selected edges from source u ;
3. the p_u sources of selected missing B-pairs oriented into u .

For family 2, if $u_j \rightarrow z$ is another selected edge at u , then z differs from w and $u_i \rightarrow w$ must dominate z ; because uz is missing, iz is an H-edge.

For family 3, let a missing pair be oriented $y \rightarrow u$. The source y cannot equal w , because $u_i \rightarrow w$ already orients the missing pair $\{u, w\}$ as $u \rightarrow w$. Since uy is missing and $u_i \rightarrow w$ dominates y , iy is an H-edge.

Families 2 and 3 cannot collide: a collision would orient the same missing unordered pair $\{u, z\}$ both as $u \rightarrow z$ and $z \rightarrow u$.

Therefore label i has at least

$$1 + (q_u - 1) + p_u = q_u + p_u$$

B-neighbours, giving

$$R_i + x_i \geq q_u + p_u. \quad (8.1)$$

A.9. Residual activity for positive surplus

Assume from now on

$$t > 0.$$

We prove

$$\rho_u \geq 1 \text{ for every } u \text{ in } B. \quad (9.1)$$

Suppose instead $\rho_u = 0$. Put

$$U = N_A(u),$$

$$T = A \setminus U.$$

Every A-B edge incident with u is selected.

There is no F-edge between U and T . Indeed, if $i \in U$, $j \in T$, and $ij \in F$, then the selected edge at u_i has its unique exception in B and must dominate j . Since ij is absent, this forces $uj \in H$, contradicting $j \in T$.

Now count residual cross-edges forced by F .

Take an F-edge ij inside U . The selected edges from u to i and j have distinct supplements, say $u_i \rightarrow w_i$ and $u_j \rightarrow w_j$. Because $w_i \neq w_j$, the selected pair $\{u, j\}$ must dominate w_i ; since uw_i is missing, jw_i is an H-edge. Similarly iw_j is an H-edge. Both are residual: jw_i has endpoints that jointly miss the A-vertex i , and iw_j has endpoints that jointly miss j . Distinct ordered F-edge endpoints give distinct residual cross-edges. Thus $F[U]$ forces $2e(F[U])$ residual edges with A-endpoint in U .

Now take an F-edge ij inside T . Adding ij to H creates a new adjacent total-dominating pair using i or j . It cannot be $\{i, j\}$ because both miss u . Its auxiliary cannot be v : v already neighbours both i and j , so adding ij cannot repair a failure at the opposite endpoint. It cannot be u because u is adjacent to neither endpoint.

If the auxiliary z lay in A , domination of u would force $z \in U$. But the unique exception before adding ij is the other endpoint in T , so z would miss that T -vertex, creating an F-edge between U and T , impossible. Hence the auxiliary lies in B .

The resulting cross quasi-edge has its unique exception in A . Selected representatives have their unique exception in B , so this cross-edge is residual. The cross-edge together with its unique A-exception recovers the original $F[T]$ -edge, giving an injection from $F[T]$ to residual edges with A-endpoint in T .

The U - and T -families are disjoint. Therefore

$$\begin{aligned} r &\geq 2e(F[U]) + e(F[T]) \\ &\geq e(F[U]) + e(F[T]) \\ &= e(F) \\ &= r + t, \end{aligned}$$

contradicting $t > 0$.

Thus (9.1) holds, and

$$r \geq b. \quad (9.2)$$

A.11. Threshold-capacity inequality

Fix an integer $h \geq 1$ and define

$$\begin{aligned} I_h &= \{i : s_i \geq h\}, \\ W_h &= \sum_{i \in I_h} s_i, \\ Z_h &= \{u : \rho_u \geq h\}, \\ z_h &= |Z_h|. \end{aligned}$$

Choose s_i actual selected incidences for every $i \in I_h$. By (6.2), every chosen source lies in Z_h .

Call an actual selected incidence **heavy** if its label lies in I_h . Let ℓ_u be the number of heavy selected incidences from source u . Then

$$W_h \leq \sum_{u \in Z_h} \ell_u. \quad (11.1)$$

Let

$$\begin{aligned} J &= \{u \in Z_h : \ell_u \geq h\}, \\ j &= |J|. \end{aligned}$$

Fix $u \in J$ and a heavy selected edge $u_i \rightarrow w$. For each other heavy selected label k at u , $\{u, k\}$ must dominate w ; since uw is missing, kw is an H-edge. There are $\ell_u - 1 \geq h$ such distinct heavy labels.

We claim $w \in Z_h$. If none of those forced edges kw is selected from source w , all are residual and $\rho_w \geq h$. If at least one is selected from w , its heavy label k has $s_k \geq h$, and (6.2) applied to that selected incidence gives $\rho_w \geq s_k \geq h$.

Thus every supplement of a heavy selected edge from J lies in Z_h . By injectivity on missing unordered B-pairs, the number of heavy selected arcs from J is at most the number of unordered pairs inside Z_h incident with J :

$$j(z_h - j) + C(j, 2) = j \cdot z_h - j(j+1)/2. \quad (11.2)$$

Every source in $Z_h \setminus J$ contributes at most h heavy incidences. Therefore

$$W_h \leq (z_h - j)h + j \cdot z_h - j(j+1)/2. \quad (11.3)$$

If $W_h > 0$, some label needs at least h distinct selected sources in Z_h , so $z_h \geq h$. Put $q = z_h - h$.

Then

$$\begin{aligned} [h \cdot z_h + C(q, 2)] - \text{RHS}(11.3) \\ = (q-j)(q-j-1)/2 \\ \geq 0, \end{aligned} \quad (11.4)$$

because $q-j$ is an integer and the product of consecutive integers is nonnegative. Hence

$$W_h \leq h \cdot z_h + C(z_h - h, 2),$$

or equivalently

$$2W_h \leq z_h^2 - z_h + h(h+1). \quad (11.5)$$

This is a necessary capacity bound; sufficiency is not claimed.

Residual activity also gives, for $h \geq 2$,

$$r \geq b + z_h(h-1). \quad (11.6)$$

Thus any valid upper bound $r \leq r_{\max}$ implies

$$z_h \leq \text{floor}((r_{\max} - b)/(h-1)). \quad (11.7)$$

Appendix B. A source-independent twelve-label tail bound

B.1. Remove the source cutoff from the definition

For each integer $h \geq 2$, set

$$C_h(z) = \frac{z(z-1) + h(h+1)}{2}, \quad \gamma_h(0) = 0,$$

and, for $W > 0$, define $\gamma_h(W)$ as the least **integer** $z \geq h$, with no upper cutoff, for which $W \leq C_h(z)$. This minimum always exists.

For twelve demands $0 \leq s_i \leq 11$, put

$$N_h = \#\{i : s_i \geq h\}, \quad W_h = \sum_{s_i \geq h} s_i, \quad p = N_1,$$

$$D = \sum_{h=2}^{11} (N_h - \gamma_h(W_h)), \quad Q = \sum_i s_i - \sum_{h=2}^{11} \gamma_h(W_h) = p + D.$$

We prove $D \leq 6$, hence **$Q \leq 18$** . The source-count cutoff is absent from this definition. The proof below uses only lower bounds on γ_h and small capacity values; no step needs $z \leq 16$.

B.2. Clip high demands to six

For $h \geq 7$, let $N = N_h$. If $0 < N < h$, then $\gamma_h(N) = h > N$. Otherwise write $N = h + e$, with $0 \leq e \leq 12 - h \leq 5$. Since $W_h \geq hN$, a positive deficit would imply $hN \leq C_h(N-1)$. But

$$2hN - 2C_h(N-1) = -e^2 + 3e + 2h - 2 > 0.$$

The concave quadratic is positive at both endpoints of $[0, 5]$, where it has values $2h-2$ and $2h-12$. Thus every deficit at level at least seven is nonpositive.

Clipping all demands above six to six removes these nonpositive deficits. At lower levels it keeps N_h fixed and decreases W_h , so it cannot decrease D .

B.3. Clip six, five and four

When the maximum is six, let k count the sixes. Their level-six deficit is nonpositive for $k \leq 10$. For $k=11, 12$ it equals one. If l counts existing fives, the required payments are:

k, l	W5 before	W5 after	γ_5 drop
11, 0	66	55	1
11, 1	71	60	2
12, 0	72	60	2

Each drop pays the lost positive level-six deficit. Thus six-to-five clipping never decreases D .

At maximum five, the level-five deficit is nonpositive for $k \leq 9$, and equals one for $k=10, 11$. If l counts fours, the five possibilities $(k, l) = (10, 0), (10, 1), (10, 2), (11, 0), (11, 1)$ all have

$$\gamma_4(5k + 4l) - \gamma_4(4(k + l)) = 1.$$

This pays the lost deficit. The exceptional vector (5^{12}) is discharged directly: its thresholds $(\gamma_2, \gamma_3, \gamma_4, \gamma_5) = (12, 11, 11, 10)$ give $D=4$. Thus every vector is either already bounded or clips to maximum four without decreasing D .

At maximum four, the level-four deficit is nonpositive for $k \leq 8$, one for $k=9, 10$, and two for $k=11, 12$. Let l count threes. For $k=9$, the γ_3 drops at $l=0, 1, 2, 3$ are 2, 1, 1, 1; for $k=10$, the drops at $l=0, 1, 2$ are 1, 2, 1. These pay the deficit one.

For $k=11$ with the last entry at most two, γ_3 drops by two. If the last entry is three, both γ_3 and γ_2 drop from ten to nine. For (4^{12}) , both again drop from ten to nine. These pay the deficit two. Clipping to maximum three therefore cannot decrease D .

All these statements are direct substitutions in $C_h(z)$. Nonpositive-deficit cutoffs can also be checked by testing $z=k-1$, exactly as in Section B.2. Lower levels not used for payment only improve.

B.4. Two tail counts finish the bound

At maximum three, set $x=N_2$, $y=N_3$, so $0 \leq y \leq x \leq 12$. Then

$$D = x + y - \gamma_2(2x + y) - \gamma_3(3y).$$

The following three ranges suffice.

Range	$x - \gamma_2(2x+y)$ at most	$y - \gamma_3(3y)$ at most	D at most
$0 \leq y \leq 6$	5	0	5
$7 \leq y \leq 10$	4	2	6
$11 \leq y \leq 12$	3	3	6

For completeness, the strict capacity defects proving the entries are:

- For the first range, $x \leq 7$ is immediate from $\gamma_2 \geq 2$ when $x > 0$; for $8 \leq x \leq 12$, testing $z=x-6$ gives $2x - C_2(x-6) = (-x^2 + 17x - 48)/2 > 0$. The y -bound is immediate for $y \leq 3$; for $4 \leq y \leq 6$, test $z=y-1$ to get $3y - C_3(y-1) = (-y^2 + 9y - 14)/2 > 0$.
- For $7 \leq y \leq 10$, test $z=y-3$ in γ_3 : the defect is $(-y^2 + 13y - 24)/2 > 0$. Test $z=x-5$ in γ_2 : the defect is $(-x^2 + 15x + 2y - 36)/2 \geq (-x^2 + 15x - 22)/2 > 0$ for $7 \leq x \leq 12$.
- For $11 \leq y \leq 12$, test $z=y-4$ in γ_3 , obtaining $(-y^2 + 15y - 32)/2 > 0$. Only $x=11, 12$ are possible; testing $z=x-4$ in γ_2 gives $(-x^2 + 13x + 2y - 26)/2 \geq (-x^2 + 13x - 4)/2 > 0$.

The empty tails have γ equal to zero and are handled separately in the first range. Positivity over each listed interval follows by checking the endpoints of the concave quadratics. Therefore $D \leq 6$ and $Q \leq p+6 \leq 18$. QED.

B.5. Transfer to actual graphs with any source count

Use the canonical parameterized bridge (Appendix A). Let

$$a = n - 1 - \Delta, \quad b = \Delta, \quad t = m - b(a + 1) > 0.$$

An actual graph gives positive residual degrees on all b sources and demands $0 \leq s_i \leq a-1$. Its residual tails z_h satisfy $z_h \geq \gamma_h(W_h)$ whenever the demand tail is nonempty. For empty demand tails, the right side is zero. Extend these zeros to all residual levels. The tail-sum identity and demand ledger give

$$S \geq r + 2t = b + 2t + \sum_{h=2}^a z_h \geq b + 2t + \sum_{h=2}^{a-1} \gamma_h(W_h),$$

so

$$Q \geq b + 2t. \tag{B.1}$$

There is no issue when γ would exceed b : then no actual residual tail could satisfy the necessary capacity inequality in the first place.

If $1 \leq a \leq 12$, pad the demand vector with zeros to length twelve. The score is unchanged because all added higher tails have zero demand. Section B.4 therefore gives $Q \leq 18$, regardless of b .

Consequently, if $1 \leq a \leq 12$ and $b \geq 17$, positive integer surplus would give

$$19 \leq b + 2t \leq Q \leq 18,$$

a contradiction. We obtain the candidate general corollary

$$1 \leq n - 1 - \Delta \leq 12, \quad \Delta \geq 17 \implies e(G) \leq \Delta(n - \Delta).$$

This bound is attained by the corresponding complete bipartite graph; no uniqueness assertion at this bound is made. It is a consequence of the project's candidate universal bridge and this hand lemma, not a claim of independent novelty or an unrestricted solution.

Appendix C. The complete thirteen-label classification

C.1. Statement and threshold notation

Let thirteen integers satisfy $0 \leq s_i \leq 12$. For $h \geq 1$ put

$$N_h = \#\{i : s_i \geq h\}, \quad W_h = \sum_{s_i \geq h} s_i.$$

For $h \geq 2$, define

$$C_h(z) = \frac{z(z-1) + h(h+1)}{2}, \quad g_h(0) = 0,$$

and, when $W > 0$, let $g_h(W)$ be the least z in $\{h, \dots, 16\}$ with $W \leq C_h(z)$. If no such z exists, the profile is threshold-inadmissible and is excluded immediately; equivalently assign it score minus infinity. All subsequent finite calculations have defined thresholds.

Write

$$p = N_1, \quad D = \sum_{h=2}^{12} (N_h - g_h(W_h)), \quad Q = \sum_i s_i - \sum_{h=2}^{12} g_h(W_h) = p + D. \quad (\text{C.1})$$

The last identity is the Ferrers-tail sum. The established graph bridge at $t = m - 224 = 1$ requires $Q \geq 16 + 2t = 18$.

Classification. Exactly 100 demand multisets have $Q \geq 18$. Seventy have maximum demand at most four and appear in Table 1. The other thirty contain fives and appear in Table 2. None contains a demand at least six. Their scores are distributed as follows.

Q	Number of profiles
18	64
19	29
20	6
21	1

In particular $Q \leq 21$, and equality holds only for (3^{13}) .

C.2. Monotone clipping, including zeros and ones

The following clipping arguments apply to arbitrary demands, including zeros and ones.

For $h \geq 8$, let $N = N_h$. If $0 < N < h$, then $g_h \geq h > N$. Otherwise write $N = h + e$, where $0 \leq e \leq 13 - h \leq 5$. Because $W_h \geq hN$, a positive deficit would require $hN \leq C_h(N-1)$. But

$$2hN - 2C_h(N-1) = -e^2 + 3e + 2h - 2 > 0. \quad (\text{C.2})$$

This concave quadratic is positive at both endpoints of $[0, 5]$: the endpoint values are $2h-2$ and $2h-12$. Thus every deficit at level at least eight is nonpositive. Clipping demands above seven to seven removes only nonpositive deficits; at lower levels it leaves N_h fixed and decreases W_h . It cannot decrease D or Q , and preserves p .

Now clip seven to six. If the number k of sevens is at most twelve, its level-seven deficit is nonpositive: for $k \geq 7$, the same capacity defect is $12 + 3e - e^2 > 0$ with $e = k - 7 \leq 5$. The only positive case is (7^{13}) , whose level-seven deficit is one. The level-six threshold drops from $g_6(91) = 13$ to $g_6(78) = 12$, paying that unit. All lower deficits improve or stay fixed.

For clipping six to five, the level-six deficit is nonpositive unless $k=11, 12, 13$; in those three cases it is one. Let l count the fives already present. The complete positive-deficit payment table is:

k	l	Drop in g_5
11	0	1
11	1	2
11	2	1
12	0	2
12	1	1
13	0	1

Each drop is $g_5(6k+5l) - g_5(5(k+l))$, so it pays the lost level-six deficit.

Finally clip five to four. The level-five deficit is nonpositive for $k \leq 9$, one for $k=10, 11$, and two for $k=12, 13$. If l counts the existing fours, then $g_4(5k+4l) - g_4(4(k+l)) = 1$ in every positive-deficit case: $l=0, \dots, 3$ for $k=10$, $l=0, \dots, 2$ for $k=11$, $l=0, 1$ for $k=12$, and $l=0$ for $k=13$.

When another unit is needed, it comes from level three:

k,l	W3 before	W3 after	Guaranteed drop in g_3
12,0	60 or 63	48 or 51	1
12,1	64	52	1
13,0	65	52	1

In the first row the remaining entry is any of $0, 1, 2, 3$; it contributes three precisely when it is three. This explicitly covers the low-demand cases absent from the earlier endpoint assumptions.

Consequently, clipping an arbitrary demand vector to four by this sequence preserves p and never decreases Q . It also preserves N_2 and N_3 . Completeness requires examining the reverse lifts; Sections C.5-C.6 do so.

C.3. Only four low-dimensional cases remain at cap four

At cap four put $x=N_2$, $y=N_3$, $z=N_4$. Then

$$0 \leq z \leq y \leq x \leq p \leq 13, \quad D_4(x, y, z) = x + y + z - g_2(2x + y + z) - g_3(3y + z) - g_4(4z). \quad (C.3)$$

The multiset is $(0^{(13-p)}, 1^{(p-x)}, 2^{(x-y)}, 3^{(y-z)}, 4^z)$.

For $x \geq 1$, adjoining a demand two to the part counted by x cannot decrease D_4 : it increases x by one and W_2 by two, and increases g_2 by at most one. Indeed $C_2(z+1) - C_2(z) = z \geq 2$ once $g_2 \geq 2$. Thus when $1 \leq x \leq 10$ it suffices to bound $D_4(10, y, z)$; $x=0$ has deficit zero.

At $z=0$, the deficits for $y=0, \dots, 10$ are

$$(3, 1, 2, 3, 3, 2, 2, 3, 3, 4, 4).$$

For $0 \leq z \leq 8$, $z - g_4(4z) \leq 0$, while increasing z cannot improve either of the first two terms in (3). Hence $D_4 \leq 4$. For $z=9, 10$, the only pairs (y, z) are $(9, 9), (10, 9), (10, 10)$, with deficits $2, 3, 2$. We have proved $x \leq 10 \Rightarrow D_4 \leq 4$.

If $x=11$ and $y \leq 10$, comparison with $x=10$ gives $D_4 \leq 5$. At $x=y=11$, the values for $z=0, \dots, 11$ are

$$(5, 2, 2, 3, 4, 4, 4, 3, 3, 4, 3, 4).$$

Thus $x \leq 11 \Rightarrow D_4 \leq 5$. The requirement $p + D_4 \geq 18$ now forces exactly one of

$$(p, x) = (12, 12), (13, 11), (13, 12), (13, 13). \quad (\text{C.4})$$

Indeed $p \leq 11$ gives $Q \leq 16$; $p=12, x \leq 11$ gives $Q \leq 17$; and $p=13, x \leq 10$ gives $Q \leq 17$.

C.4. The seventy cap-four profiles

Table 1 lists every permitted value of y for each z and each pair in (4). A dash means the set is empty. The inequalities $z \leq y \leq x$ are always imposed.

z	$p=12, x=12$	$p=13, x=11$	$p=13, x=12$	$p=13, x=13$
0	11,12	9,11	0,7,9,10,11,12	0,3,4,5,7,8,9,10,11,12,13
1	-	-	-	-
2	-	-	-	13
3	-	-	12	10,12,13
4	-	-	10,11,12	9,10,11,12,13
5	-	-	12	11,12,13
6	-	-	12	11,12,13
7	-	-	-	11,12,13
8	-	-	-	11,12,13
9	-	-	11,12	10,11,12,13
10	-	-	12	10,11,12,13
11	12	-	11,12	11,12,13
12	12	-	12	12,13
13	-	-	-	13
Count	4	2	18	46

Here is a direct method to verify both inclusion and omission in this table, without a demand search. Fix p, x, z ; substitute (3) into $p + D_4 \geq 18$. The only changing thresholds are $g_2(2x+y+z)$ and $g_3(3y+z)$. Their changes occur just after

$$y = C_2(a) - 2x - z, \quad y = \left\lfloor \frac{C_3(b) - z}{3} \right\rfloor. \quad (\text{C.5})$$

Within each resulting integer interval both thresholds are constant, so Q is y plus a fixed integer. Intersect that interval with the resulting lower bound on y . Check $y=z=0$ separately, since $g_3(0)=0$. This produces exactly Table 1.

For example, at $p=13, x=12, z=0$, (3) gives

$$Q = 25 + y - g_2(24 + y) - g_3(3y).$$

For $y=0, \dots, 12$, its values are

$$(18, 15, 16, 17, 17, 17, 17, 18, 17, 18, 18, 19, 19),$$

yielding the indicated six values of y . Each other row uses precisely the same two threshold boundaries. The capacity values needed for this table and the lifts below are:

z	$C_2(z)$	$C_3(z)$	$C_4(z)$	$C_5(z)$	$C_6(z)$
2	4	-	-	-	-
3	6	9	-	-	-
4	9	12	16	-	-
5	13	16	20	25	-
6	18	21	25	30	36
7	24	27	31	36	42
8	31	34	38	43	49
9	39	42	46	51	57
10	48	51	55	60	66
11	58	61	65	70	76
12	69	72	76	81	87

z	C2(z)	C3(z)	C4(z)	C5(z)	C6(z)
13	81	84	88	93	99

All threshold arguments in Sections C.3-C.6 are at most 78, so these capacities suffice; no threshold above thirteen is needed there.

C.5. All cap-five preimages: thirty more profiles

Any cap-five profile with $Q \geq 18$ clips to Table 1. To recover it, replace k of the z fours by fives, where $1 \leq k \leq z$. Its exact score is

$$\begin{aligned}
Q_5(p, x, y, z, k) = & p + x + y + z + k \\
& - g_2(2x + y + z + k) - g_3(3y + z + k) \\
& - g_4(4z + k) - g_5(5k).
\end{aligned} \tag{C.6}$$

The complete solutions are Table 2. All have $p=x=13$.

y	z	Permitted k	Number
12	10	5	1
12	12	5,6,7,12	4
13	10	5,6	2
13	11	5,6,7,8,10,11	6
13	12	4,5,6,7,10,12	6
13	13	3,4,5,6,7,8,9,10,11,12,13	11
Total			30

The corresponding multiset is $(2^{(13-y)}, 3^{(y-z)}, 4^{(z-k)}, 5^k)$.

To check completeness, apply the same interval argument to (6). Its threshold changes occur just after

$$\begin{aligned}
k &= C_2(a) - (2x + y + z), \\
k &= C_3(b) - (3y + z), \\
k &= C_4(c) - 4z, \\
k &= \lfloor C_5(d)/5 \rfloor.
\end{aligned} \tag{C.7}$$

Between changes, $Q_5 = k + \text{constant}$; intersect with $Q_5 \geq 18$. Table 1 has altogether 366 possible positive lifts, split by (7) into 230 affine intervals. The complete preimage appendix (Appendix D) prints every interval, its constant threshold vector, the formula for Q_5 , and its accepted integers. It includes rejected lifts from **every** Table 1 row with $z > 0$, including those with a zero or a one. Thus Table 2 is not an inference from the historical survivor list.

C.6. No original profile containing six or more is hidden by clipping

Suppose an original profile with $Q \geq 18$ contains a value at least six. After clipping to six it still has $Q \geq 18$ and contains a six. After the next clipping, it is a Table 2 profile containing fives. Therefore it suffices to lift one of the thirty profiles in Table 2: raise j of its k fives to six, where $1 \leq j \leq k$.

The exact score of that lift is

$$\begin{aligned}
Q_6(y, z, k, j) = & 26 + y + z + k + j \\
& - g_2(26 + y + z + k + j) - g_3(3y + z + k + j) \\
& - g_4(4z + k + j) - g_5(5k + j) - g_6(6j).
\end{aligned} \tag{C.8}$$

Again, the score is $j + \text{constant}$ between threshold changes. The changes occur just after the four capacities minus their displayed bases, and after $\text{floor}(C_6(d)/6)$. The maximum on each interval is its value at the right endpoint. This gives:

y	z	k ranges from Table 2	Maximum Q6 over all permitted k and $1 \leq j \leq k$
12	10	5	14
12	12	5,6,7,12	15
13	10	5,6	15
13	11	5,6,7,8,10,11	15
13	12	4,5,6,7,10,12	16
13	13	3,4,5,6,7,8,9,10,11,12,13	16

For example, lifting (5^{13}) gives

$$Q_6 = 65 + j - g_2(65 + j) - g_3(65 + j) - g_4(65 + j) - g_5(65 + j) - g_6(6j).$$

At $j=1, \dots, 13$, these scores are

$$(13, 14, 15, 16, 16, 16, 16, 15, 15, 15, 16, 15, 15).$$

There are only 225 positive six-lifts in total, divided into 131 affine intervals. Every interval and its maximum is printed in the second half of the appendix. All have score at most sixteen, contradicting $Q \geq 18$.

This rules out all original demands at least six, including seven through twelve: the forward clipping already proved that none could disappear without leaving a six-lift with a score at least as large. Together with Sections C.4-C.5, this proves the classification. QED.

Appendix D. Complete preimage arithmetic

Companion to the written hand classification. Each interval records the affine formula $Q(v)=\text{offset}+v$; threshold vectors are constant throughout it. The final point therefore attains the interval maximum. All omitted accepted-value sets below are empty.

Cap-five lifts of every cap-four profile containing a four

p,x,y,z	k interval	(g2,g3,g4,g5)	Q(k)	accepted k
12,12,12,11	1-1	10,10,9,5	13+k	none
12,12,12,11	2-2	11,10,9,5	12+k	none
12,12,12,11	3-4	11,10,10,5	11+k	none
12,12,12,11	5-5	11,11,10,5	10+k	none
12,12,12,11	6-6	11,11,10,6	9+k	none
12,12,12,11	7-7	11,11,10,7	8+k	none
12,12,12,11	8-8	11,11,10,8	7+k	none
12,12,12,11	9-10	11,11,10,9	6+k	none
12,12,12,11	11-11	11,11,10,10	5+k	none
12,12,12,12	1-3	11,10,10,5	12+k	none
12,12,12,12	4-5	11,11,10,5	11+k	none
12,12,12,12	6-6	11,11,10,6	10+k	none
12,12,12,12	7-7	11,11,10,7	9+k	none
12,12,12,12	8-8	11,11,11,8	7+k	none
12,12,12,12	9-10	11,11,11,9	6+k	none
12,12,12,12	11-12	12,11,11,10	4+k	none
13,12,12,3	1-3	10,9,4,5	12+k	none
13,12,10,4	1-1	9,9,5,5	11+k	none
13,12,10,4	2-4	10,9,5,5	10+k	none
13,12,11,4	1-4	10,9,5,5	11+k	none
13,12,12,4	1-2	10,9,5,5	12+k	none
13,12,12,4	3-4	10,10,5,5	11+k	none
13,12,12,5	1-1	10,9,6,5	12+k	none
13,12,12,5	2-5	10,10,6,5	11+k	none
13,12,12,6	1-1	10,10,6,5	12+k	none
13,12,12,6	2-5	10,10,7,5	11+k	none
13,12,12,6	6-6	10,10,7,6	10+k	none
13,12,11,9	1-2	10,10,8,5	12+k	none
13,12,11,9	3-4	10,10,9,5	11+k	none
13,12,11,9	5-5	11,10,9,5	10+k	none
13,12,11,9	6-6	11,10,9,6	9+k	none
13,12,11,9	7-7	11,10,9,7	8+k	none
13,12,11,9	8-8	11,10,9,8	7+k	none
13,12,11,9	9-9	11,10,9,9	6+k	none
13,12,12,9	1-2	10,10,8,5	13+k	none
13,12,12,9	3-3	10,10,9,5	12+k	none
13,12,12,9	4-5	11,10,9,5	11+k	none
13,12,12,9	6-6	11,10,9,6	10+k	none
13,12,12,9	7-7	11,11,9,7	8+k	none
13,12,12,9	8-8	11,11,9,8	7+k	none
13,12,12,9	9-9	11,11,9,9	6+k	none
13,12,12,10	1-2	10,10,9,5	13+k	none
13,12,12,10	3-5	11,10,9,5	12+k	none
13,12,12,10	6-6	11,11,9,6	10+k	none
13,12,12,10	7-7	11,11,10,7	8+k	none
13,12,12,10	8-8	11,11,10,8	7+k	none
13,12,12,10	9-10	11,11,10,9	6+k	none
13,12,11,11	1-2	10,10,9,5	13+k	none
13,12,11,11	3-5	11,10,10,5	11+k	none
13,12,11,11	6-6	11,10,10,6	10+k	none
13,12,11,11	7-7	11,10,10,7	9+k	none
13,12,11,11	8-8	11,11,10,8	7+k	none
13,12,11,11	9-10	11,11,10,9	6+k	none
13,12,11,11	11-11	11,11,10,10	5+k	none
13,12,12,11	1-1	10,10,9,5	14+k	none
13,12,12,11	2-2	11,10,9,5	13+k	none
13,12,12,11	3-4	11,10,10,5	12+k	none
13,12,12,11	5-5	11,11,10,5	11+k	none
13,12,12,11	6-6	11,11,10,6	10+k	none
13,12,12,11	7-7	11,11,10,7	9+k	none
13,12,12,11	8-8	11,11,10,8	8+k	none
13,12,12,11	9-10	11,11,10,9	7+k	none

p,x,y,z	k interval	(g2,g3,g4,g5)	Q(k)	accepted k
13,12,12,11	11-11	11,11,10,10	6+k	none
13,12,12,12	1-3	11,10,10,5	13+k	none
13,12,12,12	4-5	11,11,10,5	12+k	none
13,12,12,12	6-6	11,11,10,6	11+k	none
13,12,12,12	7-7	11,11,10,7	10+k	none
13,12,12,12	8-8	11,11,11,8	8+k	none
13,12,12,12	9-10	11,11,11,9	7+k	none
13,12,12,12	11-12	12,11,11,10	5+k	none
13,13,13,2	1-1	10,9,4,5	13+k	none
13,13,13,2	2-2	10,10,4,5	12+k	none
13,13,10,3	1-1	10,8,4,5	12+k	none
13,13,10,3	2-3	10,9,4,5	11+k	none
13,13,12,3	1-3	10,9,4,5	13+k	none
13,13,13,3	1-3	10,10,4,5	13+k	none
13,13,9,4	1-3	10,8,5,5	11+k	none
13,13,9,4	4-4	10,9,5,5	10+k	none
13,13,10,4	1-4	10,9,5,5	11+k	none
13,13,11,4	1-4	10,9,5,5	12+k	none
13,13,12,4	1-2	10,9,5,5	13+k	none
13,13,12,4	3-4	10,10,5,5	12+k	none
13,13,13,4	1-4	10,10,5,5	13+k	none
13,13,11,5	1-4	10,9,6,5	12+k	none
13,13,11,5	5-5	10,10,6,5	11+k	none
13,13,12,5	1-1	10,9,6,5	13+k	none
13,13,12,5	2-5	10,10,6,5	12+k	none
13,13,13,5	1-4	10,10,6,5	13+k	none
13,13,13,5	5-5	11,10,6,5	12+k	none
13,13,11,6	1-1	10,9,6,5	13+k	none
13,13,11,6	2-3	10,9,7,5	12+k	none
13,13,11,6	4-5	10,10,7,5	11+k	none
13,13,11,6	6-6	11,10,7,6	9+k	none
13,13,12,6	1-1	10,10,6,5	13+k	none
13,13,12,6	2-4	10,10,7,5	12+k	none
13,13,12,6	5-5	11,10,7,5	11+k	none
13,13,12,6	6-6	11,10,7,6	10+k	none
13,13,13,6	1-1	10,10,6,5	14+k	none
13,13,13,6	2-3	10,10,7,5	13+k	none
13,13,13,6	4-5	11,10,7,5	12+k	none
13,13,13,6	6-6	11,10,7,6	11+k	none
13,13,11,7	1-2	10,9,7,5	13+k	none
13,13,11,7	3-3	10,10,7,5	12+k	none
13,13,11,7	4-4	10,10,8,5	11+k	none
13,13,11,7	5-5	11,10,8,5	10+k	none
13,13,11,7	6-6	11,10,8,6	9+k	none
13,13,11,7	7-7	11,10,8,7	8+k	none
13,13,12,7	1-3	10,10,7,5	13+k	none
13,13,12,7	4-5	11,10,8,5	11+k	none
13,13,12,7	6-6	11,10,8,6	10+k	none
13,13,12,7	7-7	11,10,8,7	9+k	none
13,13,13,7	1-2	10,10,7,5	14+k	none
13,13,13,7	3-3	11,10,7,5	13+k	none
13,13,13,7	4-5	11,10,8,5	12+k	none
13,13,13,7	6-6	11,11,8,6	10+k	none
13,13,13,7	7-7	11,11,8,7	9+k	none
13,13,11,8	1-1	10,9,8,5	13+k	none
13,13,11,8	2-3	10,10,8,5	12+k	none
13,13,11,8	4-5	11,10,8,5	11+k	none
13,13,11,8	6-6	11,10,8,6	10+k	none
13,13,11,8	7-7	11,10,9,7	8+k	none
13,13,11,8	8-8	11,10,9,8	7+k	none
13,13,12,8	1-2	10,10,8,5	13+k	none
13,13,12,8	3-5	11,10,8,5	12+k	none
13,13,12,8	6-6	11,10,8,6	11+k	none
13,13,12,8	7-7	11,10,9,7	9+k	none
13,13,12,8	8-8	11,11,9,8	7+k	none
13,13,13,8	1-1	10,10,8,5	14+k	none
13,13,13,8	2-4	11,10,8,5	13+k	none
13,13,13,8	5-5	11,11,8,5	12+k	none
13,13,13,8	6-6	11,11,8,6	11+k	none
13,13,13,8	7-7	11,11,9,7	9+k	none
13,13,13,8	8-8	11,11,9,8	8+k	none
13,13,10,9	1-2	10,9,8,5	13+k	none
13,13,10,9	3-3	10,9,9,5	12+k	none
13,13,10,9	4-5	11,10,9,5	10+k	none

p,x,y,z	k interval	(g2,g3,g4,g5)	Q(k)	accepted k
13,13,10,9	6-6	11,10,9,6	9+k	none
13,13,10,9	7-7	11,10,9,7	8+k	none
13,13,10,9	8-8	11,10,9,8	7+k	none
13,13,10,9	9-9	11,10,9,9	6+k	none
13,13,11,9	1-2	10,10,8,5	13+k	none
13,13,11,9	3-5	11,10,9,5	11+k	none
13,13,11,9	6-6	11,10,9,6	10+k	none
13,13,11,9	7-7	11,10,9,7	9+k	none
13,13,11,9	8-8	11,10,9,8	8+k	none
13,13,11,9	9-9	11,10,9,9	7+k	none
13,13,12,9	1-1	10,10,8,5	14+k	none
13,13,12,9	2-2	11,10,8,5	13+k	none
13,13,12,9	3-5	11,10,9,5	12+k	none
13,13,12,9	6-6	11,10,9,6	11+k	none
13,13,12,9	7-7	11,11,9,7	9+k	none
13,13,12,9	8-8	11,11,9,8	8+k	none
13,13,12,9	9-9	11,11,9,9	7+k	none
13,13,13,9	1-2	11,10,8,5	14+k	none
13,13,13,9	3-3	11,10,9,5	13+k	none
13,13,13,9	4-5	11,11,9,5	12+k	none
13,13,13,9	6-6	11,11,9,6	11+k	none
13,13,13,9	7-7	11,11,9,7	10+k	none
13,13,13,9	8-8	11,11,9,8	9+k	none
13,13,13,9	9-9	11,11,9,9	8+k	none
13,13,10,10	1-2	10,9,9,5	13+k	none
13,13,10,10	3-5	11,10,9,5	11+k	none
13,13,10,10	6-6	11,10,9,6	10+k	none
13,13,10,10	7-7	11,10,10,7	8+k	none
13,13,10,10	8-8	11,10,10,8	7+k	none
13,13,10,10	9-10	11,10,10,9	6+k	none
13,13,11,10	1-1	10,10,9,5	13+k	none
13,13,11,10	2-5	11,10,9,5	12+k	none
13,13,11,10	6-6	11,10,9,6	11+k	none
13,13,11,10	7-7	11,10,10,7	9+k	none
13,13,11,10	8-8	11,10,10,8	8+k	none
13,13,11,10	9-10	11,11,10,9	6+k	none
13,13,12,10	1-5	11,10,9,5	13+k	5
13,13,12,10	6-6	11,11,9,6	11+k	none
13,13,12,10	7-7	11,11,10,7	9+k	none
13,13,12,10	8-8	11,11,10,8	8+k	none
13,13,12,10	9-10	11,11,10,9	7+k	none
13,13,13,10	1-2	11,10,9,5	14+k	none
13,13,13,10	3-5	11,11,9,5	13+k	5
13,13,13,10	6-6	11,11,9,6	12+k	6
13,13,13,10	7-7	11,11,10,7	10+k	none
13,13,13,10	8-8	11,11,10,8	9+k	none
13,13,13,10	9-9	11,11,10,9	8+k	none
13,13,13,10	10-10	12,11,10,9	7+k	none
13,13,11,11	1-2	11,10,9,5	13+k	none
13,13,11,11	3-5	11,10,10,5	12+k	none
13,13,11,11	6-6	11,10,10,6	11+k	none
13,13,11,11	7-7	11,10,10,7	10+k	none
13,13,11,11	8-8	11,11,10,8	8+k	none
13,13,11,11	9-10	11,11,10,9	7+k	none
13,13,11,11	11-11	12,11,10,10	5+k	none
13,13,12,11	1-2	11,10,9,5	14+k	none
13,13,12,11	3-4	11,10,10,5	13+k	none
13,13,12,11	5-5	11,11,10,5	12+k	none
13,13,12,11	6-6	11,11,10,6	11+k	none
13,13,12,11	7-7	11,11,10,7	10+k	none
13,13,12,11	8-8	11,11,10,8	9+k	none
13,13,12,11	9-9	11,11,10,9	8+k	none
13,13,12,11	10-10	12,11,10,9	7+k	none
13,13,12,11	11-11	12,11,10,10	6+k	none
13,13,13,11	1-1	11,10,9,5	15+k	none
13,13,13,11	2-2	11,11,9,5	14+k	none
13,13,13,11	3-5	11,11,10,5	13+k	5
13,13,13,11	6-6	11,11,10,6	12+k	6
13,13,13,11	7-7	11,11,10,7	11+k	7
13,13,13,11	8-8	11,11,10,8	10+k	8
13,13,13,11	9-10	12,11,10,9	8+k	10
13,13,13,11	11-11	12,11,10,10	7+k	11
13,13,12,12	1-3	11,10,10,5	14+k	none
13,13,12,12	4-5	11,11,10,5	13+k	5

p,x,y,z	k interval	(g2,g3,g4,g5)	Q(k)	accepted k
13,13,12,12	6-6	11,11,10,6	12+k	6
13,13,12,12	7-7	11,11,10,7	11+k	7
13,13,12,12	8-8	11,11,11,8	9+k	none
13,13,12,12	9-10	12,11,11,9	7+k	none
13,13,12,12	11-12	12,11,11,10	6+k	12
13,13,13,12	1-5	11,11,10,5	14+k	4,5
13,13,13,12	6-6	11,11,10,6	13+k	6
13,13,13,12	7-7	11,11,10,7	12+k	7
13,13,13,12	8-8	12,11,11,8	9+k	none
13,13,13,12	9-10	12,11,11,9	8+k	10
13,13,13,12	11-12	12,12,11,10	6+k	12
13,13,13,13	1-3	11,11,10,5	15+k	3
13,13,13,13	4-5	11,11,11,5	14+k	4,5
13,13,13,13	6-6	11,11,11,6	13+k	6
13,13,13,13	7-7	12,11,11,7	11+k	7
13,13,13,13	8-8	12,11,11,8	10+k	8
13,13,13,13	9-9	12,11,11,9	9+k	9
13,13,13,13	10-10	12,12,11,9	8+k	10
13,13,13,13	11-12	12,12,11,10	7+k	11,12
13,13,13,13	13-13	12,12,11,11	6+k	13

Cap-six lifts of all thirty cap-five survivors

yz,k	j interval	(g2,g3,g4,g5,g6)	Q(j)	interval maximum
12,10,5	1-1	11,11,9,6,6	10+j	11
12,10,5	2-5	11,11,10,6,6	9+j	14
12,12,5	1-2	11,11,10,6,6	11+j	13
12,12,5	3-3	11,11,11,6,6	10+j	13
12,12,5	4-5	12,11,11,6,6	9+j	14
12,12,6	1-1	11,11,10,7,6	11+j	12
12,12,6	2-2	11,11,11,7,6	10+j	12
12,12,6	3-6	12,11,11,7,6	9+j	15
12,12,7	1-1	11,11,11,7,6	11+j	12
12,12,7	2-6	12,11,11,8,6	9+j	15
12,12,7	7-7	12,12,11,8,7	7+j	14
12,12,12	1-1	12,11,11,11,6	11+j	12
12,12,12	2-5	12,12,11,11,6	10+j	15
12,12,12	6-6	12,12,12,11,6	9+j	15
12,12,12	7-7	12,12,12,11,7	8+j	15
12,12,12	8-8	13,12,12,11,8	6+j	14
12,12,12	9-9	13,12,12,11,9	5+j	14
12,12,12	10-10	13,12,12,11,10	4+j	14
12,12,12	11-11	13,12,12,12,10	3+j	14
12,12,12	12-12	13,12,12,12,11	2+j	14
13,10,5	1-1	11,11,9,6,6	11+j	12
13,10,5	2-4	11,11,10,6,6	10+j	14
13,10,5	5-5	12,11,10,6,6	9+j	14
13,10,6	1-3	11,11,10,7,6	10+j	13
13,10,6	4-6	12,11,10,7,6	9+j	15
13,11,5	1-3	11,11,10,6,6	11+j	14
13,11,5	4-5	12,11,10,6,6	10+j	15
13,11,6	1-2	11,11,10,7,6	11+j	13
13,11,6	3-5	12,11,10,7,6	10+j	15
13,11,6	6-6	12,12,11,7,6	8+j	14
13,11,7	1-1	11,11,10,7,6	12+j	13
13,11,7	2-4	12,11,10,8,6	10+j	14
13,11,7	5-6	12,12,11,8,6	8+j	14
13,11,7	7-7	12,12,11,8,7	7+j	14
13,11,8	1-3	12,11,10,8,6	11+j	14
13,11,8	4-6	12,12,11,9,6	8+j	14
13,11,8	7-7	12,12,11,9,7	7+j	14
13,11,8	8-8	12,12,11,9,8	6+j	14
13,11,10	1-1	12,11,10,9,6	12+j	13
13,11,10	2-6	12,12,11,10,6	9+j	15
13,11,10	7-7	12,12,11,10,7	8+j	15
13,11,10	8-8	12,12,11,10,8	7+j	15
13,11,10	9-9	12,12,11,10,9	6+j	15
13,11,10	10-10	13,12,11,10,10	4+j	14
13,11,11	1-5	12,12,11,10,6	10+j	15
13,11,11	6-6	12,12,11,11,6	9+j	15

yz,k	j interval	(g2,g3,g4,g5,g6)	Q(j)	interval maximum
13,11,11	7-7	12,12,11,11,7	8+j	15
13,11,11	8-8	12,12,11,11,8	7+j	15
13,11,11	9-9	13,12,11,11,9	5+j	14
13,11,11	10-10	13,12,11,11,10	4+j	14
13,11,11	11-11	13,12,12,11,10	3+j	14
13,12,4	1-3	11,11,10,5,6	12+j	15
13,12,4	4-4	12,11,11,5,6	10+j	14
13,12,5	1-2	11,11,10,6,6	12+j	14
13,12,5	3-5	12,11,11,6,6	10+j	15
13,12,6	1-1	11,11,10,7,6	12+j	13
13,12,6	2-4	12,11,11,7,6	10+j	14
13,12,6	5-6	12,12,11,7,6	9+j	15
13,12,7	1-1	12,11,11,7,6	11+j	12
13,12,7	2-3	12,11,11,8,6	10+j	13
13,12,7	4-6	12,12,11,8,6	9+j	15
13,12,7	7-7	12,12,11,8,7	8+j	15
13,12,10	1-1	12,12,11,9,6	11+j	12
13,12,10	2-6	12,12,11,10,6	10+j	16
13,12,10	7-7	12,12,11,10,7	9+j	16
13,12,10	8-8	12,12,12,10,8	7+j	15
13,12,10	9-9	13,12,12,10,9	5+j	14
13,12,10	10-10	13,12,12,10,10	4+j	14
13,12,12	1-5	12,12,11,11,6	11+j	16
13,12,12	6-6	12,12,12,11,6	10+j	16
13,12,12	7-7	13,12,12,11,7	8+j	15
13,12,12	8-8	13,12,12,11,8	7+j	15
13,12,12	9-9	13,12,12,11,9	6+j	15
13,12,12	10-10	13,13,12,11,10	4+j	14
13,12,12	11-11	13,13,12,12,10	3+j	14
13,12,12	12-12	13,13,12,12,11	2+j	14
13,13,3	1-3	11,11,11,5,6	11+j	14
13,13,4	1-2	11,11,11,5,6	12+j	14
13,13,4	3-4	12,11,11,5,6	11+j	15
13,13,5	1-1	11,11,11,6,6	12+j	13
13,13,5	2-4	12,11,11,6,6	11+j	15
13,13,5	5-5	12,12,11,6,6	10+j	15
13,13,6	1-3	12,11,11,7,6	11+j	14
13,13,6	4-6	12,12,11,7,6	10+j	16
13,13,7	1-1	12,11,11,7,6	12+j	13
13,13,7	2-2	12,11,11,8,6	11+j	13
13,13,7	3-6	12,12,11,8,6	10+j	16
13,13,7	7-7	12,12,12,8,7	8+j	15
13,13,8	1-1	12,11,11,8,6	12+j	13
13,13,8	2-3	12,12,11,8,6	11+j	14
13,13,8	4-5	12,12,11,9,6	10+j	15
13,13,8	6-6	12,12,12,9,6	9+j	15
13,13,8	7-7	12,12,12,9,7	8+j	15
13,13,8	8-8	12,12,12,9,8	7+j	15
13,13,9	1-4	12,12,11,9,6	11+j	15
13,13,9	5-6	12,12,12,9,6	10+j	16
13,13,9	7-7	12,12,12,10,7	8+j	15
13,13,9	8-8	12,12,12,10,8	7+j	15
13,13,9	9-9	13,12,12,10,9	5+j	14
13,13,10	1-1	12,12,11,9,6	12+j	13
13,13,10	2-3	12,12,11,10,6	11+j	14
13,13,10	4-6	12,12,12,10,6	10+j	16
13,13,10	7-7	12,12,12,10,7	9+j	16
13,13,10	8-8	13,12,12,10,8	7+j	15
13,13,10	9-9	13,12,12,10,9	6+j	15
13,13,10	10-10	13,12,12,10,10	5+j	15
13,13,11	1-2	12,12,11,10,6	12+j	14
13,13,11	3-5	12,12,12,10,6	11+j	16
13,13,11	6-6	12,12,12,11,6	10+j	16
13,13,11	7-7	13,12,12,11,7	8+j	15
13,13,11	8-8	13,12,12,11,8	7+j	15
13,13,11	9-9	13,12,12,11,9	6+j	15
13,13,11	10-11	13,13,12,11,10	4+j	15
13,13,12	1-1	12,12,11,11,6	12+j	13
13,13,12	2-5	12,12,12,11,6	11+j	16
13,13,12	6-6	13,12,12,11,6	10+j	16
13,13,12	7-7	13,12,12,11,7	9+j	16
13,13,12	8-8	13,12,12,11,8	8+j	16
13,13,12	9-9	13,13,12,11,9	6+j	15
13,13,12	10-10	13,13,12,11,10	5+j	15

y,z,k	j interval	(g2,g3,g4,g5,g6)	Q(j)	interval maximum
13,13,12	11-11	13,13,12,12,10	4+j	15
13,13,12	12-12	13,13,12,12,11	3+j	15
13,13,13	1-4	12,12,12,11,6	12+j	16
13,13,13	5-5	13,12,12,11,6	11+j	16
13,13,13	6-6	13,12,12,12,6	10+j	16
13,13,13	7-7	13,12,12,12,7	9+j	16
13,13,13	8-8	13,13,12,12,8	7+j	15
13,13,13	9-9	13,13,12,12,9	6+j	15
13,13,13	10-11	13,13,12,12,10	5+j	16
13,13,13	12-12	13,13,13,12,11	3+j	15
13,13,13	13-13	13,13,13,12,12	2+j	15

No interval in the second table reaches 18.

Appendix E. Complete residual-tail reconstruction

For $(a, b) = (13, 16)$ and $t = m - 224 > 0$, put $g_{13} = 0$ and use the thresholds $g_2, \dots, g_{12}$ from Appendix C. Every actual residual degree is in $\{1, \dots, 13\}$. Let

$$z_h = \#\{u : \rho_u \geq h\}, \quad r = 16 + \sum_{h=2}^{13} z_h.$$

The thresholds and the demand ledger give $z_h \geq g_h$ and $\lambda = S - r - 2t \geq 0$. Substitution yields

$$Q - (16 + 2t) = \lambda + \sum_{h=2}^{13} (z_h - g_h).$$

Thus the residual rows are exactly those reconstructed by assigning this many nonnegative integer units to $\lambda, z_2 - g_2, \dots, z_{13} - g_{13}$, imposing $16 \geq z_2 \geq \dots \geq z_{13} \geq 0$. This means completeness of the integer relaxation; no sufficiency for an actual graph is asserted. A monotone tail determines a unique histogram, with multiplicities $16 - z_2$ at one, $z_h - z_{h+1}$ at $h = 2, \dots, 12$, and z_{13} at thirteen. The thresholds g_h need not themselves be monotone, so monotonicity must be imposed after the slack is assigned.

At $t = 1$, at most three units are available. Applying this rule to the 100 profiles in Appendix C gives the following complete distribution.

Q	λ	Rows
18	0	64
19	0	96
19	1	29
20	0	42
20	1	18
20	2	6
21	0	9
21	1	5
21	2	2
21	3	1

The total is 272. Appendix C shows that every high-score profile containing a zero has $p = 12, x = 12$ and $Q = 18$. Such a profile has no available slack at $t = 1$. Consequently every row with positive ledger slack has every demand positive. Its definition then gives $s_i = d_i - R_i$ at each label, so

$$S = \sum_i d_i - \sum_i R_i = 2(r + t) - r = r + 2t,$$

contradicting $\lambda > 0$. This excludes exactly 61 rows. The remaining 211 rows are all printed in Appendix H. Checking that list against the reconstruction rule is a finite proof obligation; the count alone is not a completeness certificate.

At $t = 2$, only the six $Q = 20$ profiles and the unique $Q = 21$ profile remain. The former have no available slack. For $s = (3^{13})$, $g_2 = g_3 = 9$ and all higher thresholds vanish. Its one unit may be placed only in λ , in z_2 , or in z_4 ; adding it at z_3 or any higher position violates monotonicity. This gives nine rows in total. The positive-ledger-slack row is $(s, \rho) = ((3^{13}), (1^7, 3^9))$ and is impossible by the same positive-demand argument. The other eight rows are all printed in Appendix F.

For the ledger-tight rows, equality in $\sum_i \max(0, d_i - R_i) \geq \sum_i (d_i - R_i)$ implies $d_i - R_i \geq 0$ at every label. Hence $d_i = R_i + s_i$ also at zero-demand labels. This is the precise tightness fact used in the envelope argument.

Appendix F. Eight explicit inequalities at 226 edges

The reconstruction in Appendix E leaves the following eight tight rows.

F.2. Tight demand and selected excess

Every one of the remaining eight rows has

$$S=r+4.$$

Since always

$$S=\sum_i \max(0, d_i - R_i) \geq \sum_i (d_i - R_i) = r+4,$$

and every displayed demand is positive, equality forces

$$s_i = d_i - R_i \quad (2.1)$$

for every label.

Let

x_i = selected degree of label i ,

$e_i = x_i - s_i \geq 0$,

$M = \sum_i x_i = \sum_u q_u$,

$E = \sum_i e_i = M - S$.

Fix a selected incidence $ui \rightarrow w$. The universal bridge gives

$$d_i \leq \rho_u + q_u - 1 \quad (2.2)$$

$$q_u + p_u \leq R_i + x_i. \quad (2.3)$$

Substitute $d_i = R_i + s_i$ and $x_i = s_i + e_i$. Eliminating R_i, q_u gives

$$e_i \geq p_u - \rho_u + 1. \quad (2.4)$$

Also

$$p_u \leq \rho_u + 2,$$

so with

$$\alpha_u = \max(0, p_u - \rho_u + 1),$$

we have

$$0 \leq \alpha_u \leq 3. \quad (2.5)$$

The selected-incidence forcing lemma also gives

$$s_i \leq \rho_u \quad (2.6)$$

on every selected incidence. In particular, if $\rho_u < \min_i s_i$, then $q_u = 0$.

F.3. Label-side upper bound

Define

$$B_0 = \sum_u q_u \alpha_u.$$

Each of the q_u selected incidences from source u lands at a label with $e_i \geq \alpha_u$. Since $\alpha_u \leq 3$, summing incidence by incidence gives

$$B_0 \leq \sum_i x_i \min(e_i, 3). \quad (3.1)$$

For every $s \geq 0, e \geq 0$,

$$(s+e) \min(e, 3) \leq (s+3)e. \quad (3.2)$$

Indeed this is immediate for $e \leq 3$, while for $e \geq 3$ it is equivalent to $3s+3e \leq se+3e$, i.e. $s(e-3) \geq 0$.

Let

$$c = \max_i s_i + 3.$$

Then (3.1)–(3.2) give

$$B_0 \leq c \sum_i e_i = cE. \quad (3.3)$$

Thus any actual graph must satisfy $B_0 \leq cE$.

F.4. Supplement Hall tails

For $k \geq 1$, define

$$H_k = \sum_{\{u: q_u \geq k+1\}} q_u - \sum_{\{w: \rho_w + q_w \geq k\}} p_w.$$

The supplement-forcing lemma says that every selected arc from source u to supplement w satisfies

$$\rho_w + q_w \geq q_u - 1.$$

Hence every outgoing selected arc from a source with $q_u \geq k+1$ must land at a vertex counted in the second term. The selected orientations form genuine source-to-supplement arcs, so

$$H_k \leq 0 \quad (4.1)$$

for every k .

No pair-capacity inequality, cumulative-threshold LP, Farkas certificate, or selected-label-group source cap is used below.

F.5. Tiny source certificates

For a source of residual degree ρ , define

$$h_k(\rho, q, p) = q \cdot 1[q \geq k+1] - p \cdot 1[\rho + q \geq k].$$

For each residual row in the table below, choose denominator D , put $c = \max s + 3$, and use the listed nonzero integer weights z_k .

For every allowed source type

$$\begin{aligned} 0 &\leq q \leq 13 - \rho, \\ 0 &\leq p \leq \rho + 2, \\ q + p &\leq 15, \end{aligned}$$

with the additional necessary condition $q=0$ whenever $\rho < \min s$, direct integer minimisation of the displayed piecewise-linear expression gives

$$D \cdot q \cdot (\alpha - c) + \sum_k z_k h_k(\rho, q, p) \geq L_\rho. \quad (5.1)$$

The table records the exact minima L_ρ . Each is checked over the displayed finite integer source domain; the verification companion gives corroborating replay instructions.

demand s	residual histogram ρ	D	c	nonzero z_k	exact L_ρ by residual degree	δ
$(2^2, 3^{11})$	$(1^7, 2, 3^8)$	2	6	$z_2=6, z_5=z_6=z_8=1$	$L_1=0, L_2=-42, L_3=-50$	1
$(2, 3^{12})$	$(1^7, 3^9)$	2	6	$z_2=3, z_3=4, z_6=z_8=1$	$L_1=0, L_3=-50$	3
(3^{13})	$(1^7, 3^8, 4)$	2	6	$z_2=7, z_6=z_7=1$	$L_1=0, L_3=-48, L_4=-60$	12
$(3^3, 13)$	$(1^6, 2, 3^9)$	2	6	$z_2=3, z_3=4, z_6=z_8=1$	$L_1=0, L_2=-12, L_3=-50$	3
$(3^4, 4^9)$	$(1^6, 3^2, 4^8)$	3	7	$z_2=11, z_5=z_6=1, z_7=2$	$L_1=0, L_3=-90, L_4=-102$	4
$(3^2, 4^{11})$	$(1^5, 2, 3, 4^9)$	3	7	$z_2=6, z_3=z_4=3, z_6=z_7=z_8=1$	$L_1=0, L_2=-24, L_3=-90, L_4=-102$	6
$(3, 4^{12})$	$(1^5, 2, 4^{10})$	2	7	$z_2=z_3=z_5=3, z_8=1$	$L_1=0, L_2=-12, L_4=-70$	1
(4^{13})	$(1^5, 3, 4^{10})$	2	7	$z_1=1, z_2=5, z_4=z_5=z_6=z_7=1$	$L_1=-3, L_3=-30, L_4=-68$	3/2

Here

$$\delta = cS + (1/D) \sum_\rho n_\rho L_\rho,$$

where n_{ρ} is the multiplicity of residual degree ρ in the row. All eight values are strictly positive.

Now sum (5.1) over all sixteen sources. The h_k terms sum to H_k , so

$$D(B\theta - cM) + \sum_k z_k H_k \geq \sum_{\rho} n_{\rho} L_{\rho}.$$

By (4.1), every $H_k \leq 0$ and every $z_k \geq 0$. Therefore

$$D(B\theta - cM) \geq \sum_{\rho} n_{\rho} L_{\rho}.$$

Adding DcS and using $E=M-S$ gives

$$B\theta - cE \geq \delta > 0. \tag{5.2}$$

Thus

$$B\theta > cE, \tag{5.3}$$

contradicting the necessary label-side inequality (3.3).

All eight nontrivial rows are impossible.

Appendix G. Four exact monotone-envelope certificates

G.1. Exact setup and the additional individual label cap

Use the selected/residual construction with

$$a=13, b=16, t=m-224=1.$$

For labels, use demand s_i , F-degree d_i , residual degree R_i and selected degree x_i . At a source use residual degree ρ_u , selected outdegree q_u and supplement indegree p_u . Write

$$S=\sum_i s_i, r=\sum_u \rho_u=\sum_i R_i.$$

The threshold-slack reduction (Appendix E) leaves 211 rows with

$$S=r+2.$$

Since $s_i=\max(0, d_i-R_i)$ and $\sum_i (d_i-R_i)=r+2$, this equality forces

$$d_i=R_i+s_i$$

at **every** label, including labels with zero demand. Also $x_i \geq s_i$.

The bridge gives $s_i \leq \rho_u$ on every selected incidence at i . Each source can be selected at a fixed label at most once because the selected cross-edges form a subset of the edges of a simple graph. Therefore define

$$H_s = \#\{u: \rho_u \geq s\}, \\ Q_\rho = \min(13-\rho, \#\{i: s_i \leq \rho\}).$$

Every individual label of demand s and source of residual degree ρ obeys

$$\text{Label domain:} \\ 0 \leq R \leq 12-s, \\ s \leq x \leq \min(16-R, H_s).$$

$$\text{Source domain:} \\ 0 \leq q \leq Q_\rho, \\ 0 \leq p \leq \rho+2. \quad (1)$$

The label bound $R \leq 12-s$ uses only $d_F \leq 12$, since F has thirteen vertices. The stronger isolated-C bound $d_F \leq 11$ used in older models is **not** a premise here.

The supplement bound in (1) follows from the corrected degree identity

$$\deg_H(u) = (\rho_u + q_u) + (15 - q_u - p_u) = \rho_u + 15 - p_u \geq 13.$$

It is $p \leq \rho+2$, not the previously rejected $\rho+1$ tightening. Since $q \leq 13-\rho$, the rectangular source domain already ensures $q+p \leq 15$.

G.2. The monotone-incidence inequality

For a label state define

$$d=R+s, v=16-R-x.$$

For a source state define

$$\alpha=\rho+q-1, w=16-q-p.$$

On every selected source-label incidence the bridge gives

$$s \leq \rho, \\ d \leq \alpha, \\ v \leq w. \quad (2)$$

Consequently any coordinatewise nondecreasing function $\Phi(s, d, v)$ obeys

$$\sum_i x_i \Phi(s_i, R_i+s_i, 16-R_i-x_i) \\ \leq \sum_u q_u \Phi(\rho_u, \rho_u+q_u-1, 16-q_u-p_u). \quad (3)$$

Apply monotonicity separately to each selected edge and then sum. Every label occurs exactly x_i times and every source exactly q_u times. This is a direct count on the actual selected incidences; no converse construction from an integer relaxation is assumed.

The useful new simple term is $s*v$. Both factors in (2) are nonnegative, so

$$s*v \leq \rho*w$$

on each selected incidence. It retains how demand interacts with the remaining cross-capacity, which the previous aggregate excess estimate discarded.

G.3. Resource identities and supplement tails

The exact identities are

$$\begin{aligned} \sum_i R_i &= r, \\ \sum_i x_i &= \sum_u q_u, \\ \sum_u q_u &= \sum_u p_u. \end{aligned} \quad (4)$$

For each $j \geq 1$, let

$$T_j = \sum_{u: q_u \geq j+1} q_u - \sum_{w: \rho_w + q_w \geq j} p_w.$$

Every selected arc from a source with $q_u \geq j+1$ has a supplement satisfying $\rho_w + q_w \geq q_u - 1 \geq j$. Therefore

$$T_j \leq 0. \quad (5)$$

The selected orientations count each arc once; the supplement's total incoming degree is p_w .

G.4. Exact scalar envelopes

Choose nonnegative integer coefficients λ, c, μ, τ_j , and a nondecreasing potential Φ . Define the two finite envelopes over the entire domains in (1):

$$\begin{aligned} \text{ell}_s &= \min [\lambda R + c x \\ &\quad + x \Phi(s, R+s, 16-R-x)], \\ \text{sigma}_\rho &= \min [\mu(q-p) - c q \\ &\quad + \sum_j \tau_j (q \mathbf{1}_{[q \geq j+1]} - p \mathbf{1}_{[\rho+q \geq j]}) \\ &\quad - q \Phi(\rho, \rho+q-1, 16-q-p)]. \end{aligned} \quad (6)$$

Let n_s be the demand multiplicities and n_ρ the residual-degree multiplicities. A necessary condition for an actual graph is

$$\text{gap} := \sum_s n_s \text{ell}_s + \sum_\rho n_\rho \text{sigma}_\rho - \lambda r \leq 0. \quad (7)$$

Proof. The minima in (6) are lower bounds for the values at the actual vertex states. Sum those bounds over all labels and sources. The c and μ terms cancel by (4). The τ terms are nonpositive by (5); the potential difference is nonpositive by (3). The only remaining contribution is $\lambda \sum_i R_i = \lambda r$. This proves (7). Thus any strictly positive value is an exact contradiction. No optimizer, transport matrix or numerical infeasibility decision is a premise. QED.

G.5. Four explicit potentials

Use the monotone generators

$$\begin{aligned} V(s, d, v) &= s*v, \\ E_k(s, d, v) &= \mathbf{1}_{[d+v \geq 16-k]}, \\ J(s, d, v) &= \mathbf{1}_{[s \geq 2]}, \\ R_{\{D, V_0\}}(s, d, v) &= \mathbf{1}_{[d \geq D \text{ and } v \geq V_0]}. \end{aligned}$$

All variables lie in a nonnegative domain. Each generator is coordinatewise nondecreasing, and each coefficient below is nonnegative. At a label, $E_k=1[x-s\leq k]$; at a source, $E_k=1[p\leq \rho+k-1]$.

Certificate B1

$\lambda=150$, $c=88$, $\mu=150$,
 $\tau_2=11$, $\tau_3=107$, $\tau_5=5$,

$$\begin{aligned} \Phi = & 4V + 51E_0 + 40E_1 + 40E_2 + 150J \\ & + 91R_{\{1,13\}} + 142R_{\{1,14\}} \\ & + 36R_{\{2,10\}} + 37R_{\{2,11\}} + 61R_{\{2,12\}} \\ & + 21R_{\{4,7\}} + 21R_{\{4,8\}} + 32R_{\{4,9\}} + R_{\{4,10\}}. \end{aligned}$$

This single certificate excludes 195 of the 211 tight rows.

Certificate B2

$\lambda=15$, $c=8$, $\mu=17$,
 $\tau_2=3$, $\tau_3=4$,

$$\begin{aligned} \Phi = & 3E_0 + 6E_1 + 3E_2 + 14J + 11R_{\{1,14\}} \\ & + 3R_{\{2,7\}} + 2R_{\{2,8\}} + 5R_{\{2,9\}} \\ & + 5R_{\{2,10\}} + 5R_{\{2,11\}} + 7R_{\{2,12\}} + 8R_{\{2,13\}}. \end{aligned}$$

It excludes 79 rows in total, including 13 not excluded by B1.

Certificate B3

$\lambda=18$, $c=4$, $\mu=24$,
 $\tau_2=2$, $\tau_3=4$,

$$\begin{aligned} \Phi = & V + 4E_0 + 8E_1 + 3E_2 + 7J \\ & + 2R_{\{1,12\}} + 11R_{\{1,13\}} + 14R_{\{1,14\}} \\ & + 3R_{\{3,11\}} + R_{\{3,12\}} + R_{\{4,9\}} + 3R_{\{4,10\}}. \end{aligned}$$

It excludes 24 rows in total, including the next two rows missed by B1 and B2.

Certificate C1

$\lambda=25$, $c=28$, $\mu=0$,
 $\tau_2=22$, $\tau_4=6$,
 $\Phi=V=s*v$.

It excludes ten rows in total, including the sole row missed by B1, B2 and B3. All unlisted τ_j are zero.

Together the four certificates cover

$$195 + 13 + 2 + 1 = 211$$

distinct rows. This includes the four zero-demand rows and all 207 positive-demand rows. Four is a convenient explicit cover; **minimality is not claimed**. This count concerns a different profile domain and different potentials from the earlier fixed-potential two-template result on seven hard N30 rows.

G.6. The final exceptional row, written out

The row missed by the first three certificates is

$$\begin{aligned} s = & (2, 5^{12}), \\ \rho = & (1^4, 2, 4, 5^{10}), \\ S = & 62, \quad r = 60. \end{aligned}$$

For C1, the exact minima in (6) are

```
ell_2=112, ell_5=415,
sigma_1=0, sigma_2=-138, sigma_4=-232, sigma_5=-322.
```

Consequently

```
gap = 112 + 12*415 -138 -232 -10*322 -25*60
     = 2 > 0.
```

This contradicts (7). The complete appendix (Appendix H) gives the same type of explicit arithmetic for every one of the 211 rows; the minimum assigned gap is one.

G.7. Why the envelope tables can be checked without optimization

For each fixed label value x , the expression in (6) is linear in R between the rectangle break-points. It therefore suffices to check the endpoints

```
R=0, R=min(12-s,16-x),
R=D-s-1, D-s,
R=16-V0-x, 17-V0-x
```

for the active rectangles, retaining only values inside the allowed interval. These include the integer states immediately before and after each rectangle indicator changes. The diagonal indicators depend only on $x-s$; the $s*v$ term is linear in R for fixed x .

For each fixed source value q , the source expression is linear in p between indicator changes. It suffices to check

```
p=0, rho+2;
p=rho+k-1, rho+k for each active E_k;
p=16-q-V0, 17-q-V0 for each active rectangle whose
    source degree rho+q-1 is at least D.
```

Again retain only admissible p values. These formulas account for both sides of every discrete jump, including the endpoints $p=rho+2$ and $R=12-s$. There is no hidden minimization over fractional vertex states.

Appendix H. All 211 tight endpoint rows

Each positive gap is $\text{sum}(n_s * \text{ell_s}) + \text{sum}(n_rho * \text{sigma_rho}) - \text{lambda} * r$. Every template is evaluated on every tight row before the first successful one is assigned.

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Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	(0,2,3 ¹¹)	(1 ⁷ ,2,3 ⁸)	{0: 0, 2: 1546, 3: 2247}	{1: -469, 2: -1480, 3: -2010}	470
B1	(0,3 ¹²)	(1 ⁷ ,3 ⁹)	{0: 0, 3: 2247}	{1: -469, 3: -2010}	491
B1	(0,3,4 ¹¹)	(1 ⁶ ,3,4 ⁹)	{0: 0, 3: 2247, 4: 2756}	{1: -469, 3: -1905, 4: -2268}	682
B1	(0,4 ¹²)	(1 ⁶ ,4 ¹⁰)	{0: 0, 4: 2756}	{1: -469, 4: -2268}	678
B3	(1 ² ,2 ² ,3 ⁹)	(1 ⁸ ,2,3 ⁷)	{1: 61, 2: 156, 3: 234}	{1: -72, 2: -141, 3: -180}	5
B3	(1 ² ,3 ¹¹)	(1 ⁷ ,2,3 ⁸)	{1: 61, 3: 232}	{1: -72, 2: -134, 3: -180}	2
B2	(1,2 ¹²)	(1 ⁹ ,2 ⁷)	{1: 31, 2: 152}	{1: -51, 2: -150}	1
B2	(1,2 ⁵ ,3 ⁷)	(1 ⁸ ,2 ² ,3 ⁶)	{1: 31, 2: 152, 3: 192}	{1: -51, 2: -150, 3: -150}	77
B2	(1,2 ³ ,3 ⁹)	(1 ⁷ ,2 ² ,3 ⁷)	{1: 31, 2: 152, 3: 192}	{1: -51, 2: -148, 3: -150}	32
B2	(1,2 ² ,3 ¹⁰)	(1 ⁷ ,2,3 ⁸)	{1: 31, 2: 152, 3: 192}	{1: -51, 2: -147, 3: -150}	56
B1	(1,2 ² ,3 ⁶ ,4 ⁴)	(1 ⁷ ,2,3 ⁴ ,4 ⁴)	{1: 512, 2: 1546, 3: 2247, 4: 2756}	{1: -469, 2: -1836, 3: -2010, 4: -2268}	329
B1	(1,2,3 ¹¹)	(1 ⁷ ,2,3 ⁷ ,4)	{1: 512, 2: 1546, 3: 2247}	{1: -469, 2: -1480, 3: -2010, 4: -2268}	574
B1	(1,2,3 ¹¹)	(1 ⁷ ,3 ⁹)	{1: 512, 2: 1546, 3: 2247}	{1: -469, 3: -2010}	302
B2	(1,2,3 ¹¹)	(1 ⁶ ,2 ² ,3 ⁸)	{1: 31, 2: 152, 3: 192}	{1: -51, 2: -136, 3: -150}	7
B1	(1,2,3 ⁷ ,4 ⁴)	(1 ⁷ ,3 ⁵ ,4 ⁴)	{1: 512, 2: 1546, 3: 2247, 4: 2756}	{1: -469, 3: -2010, 4: -2268}	706
B1	(1,2,3 ² ,4 ⁹)	(1 ⁶ ,2,3,4 ⁸)	{1: 512, 2: 1546, 3: 2247, 4: 2756}	{1: -469, 2: -1480, 3: -2010, 4: -2268}	458
B1	(1,2,4 ¹¹)	(1 ⁶ ,3,4 ⁹)	{1: 512, 2: 1546, 4: 2756}	{1: -469, 3: -1905, 4: -2268}	493
B1	(1,3 ¹²)	(1 ⁷ ,3 ⁸ ,4)	{1: 512, 3: 2247}	{1: -469, 3: -2010, 4: -2268}	595
B1	(1,3 ¹²)	(1 ⁶ ,2,3 ⁹)	{1: 512, 3: 2247}	{1: -469, 2: -1321, 3: -2010}	1
B1	(1,3 ⁹ ,4 ³)	(1 ⁷ ,3 ⁵ ,4 ⁴)	{1: 512, 3: 2247, 4: 2756}	{1: -469, 3: -2010, 4: -2268}	898
B1	(1,3 ⁸ ,4 ⁴)	(1 ⁶ ,2,3 ⁵ ,4 ⁴)	{1: 512, 3: 2247, 4: 2756}	{1: -469, 2: -1321, 3: -2010, 4: -2268}	405
B1	(1,3 ⁷ ,4 ⁵)	(1 ⁶ ,2,3 ⁴ ,4 ⁵)	{1: 512, 3: 2247, 4: 2756}	{1: -469, 2: -1321, 3: -2010, 4: -2268}	506
B1	(1,3 ⁶ ,4 ⁶)	(1 ⁶ ,2,3 ³ ,4 ⁶)	{1: 512, 3: 2247, 4: 2756}	{1: -469, 2: -1321, 3: -2010, 4: -2268}	607
B1	(1,3 ³ ,4 ⁹)	(1 ⁶ ,3 ² ,4 ⁸)	{1: 512, 3: 2247, 4: 2756}	{1: -469, 3: -2010, 4: -2268}	479

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	$(1,3^2,4^{10})$	$(1^6,3,4^9)$	{1: 512, 3: 2247, 4: 2756}	{1: -469, 3: -2010, 4: -2268}	580
B1	$(1,3,4^{11})$	$(1^6,3,4^{8,5})$	{1: 512, 3: 2247, 4: 2756}	{1: -469, 3: -1905, 4: -2268, 5: -2502}	810
B1	$(1,3,4^{11})$	$(1^6,4^{10})$	{1: 512, 3: 2247, 4: 2756}	{1: -469, 4: -2268}	681
B1	$(1,3,4^{11})$	$(1^5,2,3,4^9)$	{1: 512, 3: 2247, 4: 2756}	{1: -469, 2: -1321, 3: -1905, 4: -2268}	192
B1	$(1,4^{12})$	$(1^6,4^9,5)$	{1: 512, 4: 2756}	{1: -469, 4: -2268, 5: -2502}	806
B1	$(1,4^{12})$	$(1^5,2,4^{10})$	{1: 512, 4: 2756}	{1: -469, 2: -1321, 4: -2268}	188
B2	(2^{13})	$(1^8,2^8)$	{2: 152}	{1: -51, 2: -150}	8
B2	$(2^{10},3^3)$	$(1^8,2^5,3^3)$	{2: 152, 3: 192}	{1: -51, 2: -150, 3: -150}	83
B2	$(2^9,3^4)$	$(1^8,2^4,3^4)$	{2: 152, 3: 192}	{1: -51, 2: -150, 3: -150}	108
B1	$(2^8,3^5)$	$(1^8,2^3,3^5)$	{2: 1546, 3: 2247}	{1: -450, 2: -1836, 3: -2010}	95
B2	$(2^6,3^7)$	$(1^7,2^3,3^6)$	{2: 152, 3: 192}	{1: -51, 2: -150, 3: -150}	84
B1	$(2^5,3^8)$	$(1^7,2^2,3^7)$	{2: 1546, 3: 2247}	{1: -450, 2: -1836, 3: -2010}	14
B1	$(2^4,3^9)$	$(1^7,2^2,3^6,4)$	{2: 1546, 3: 2247}	{1: -450, 2: -1836, 3: -2010, 4: -2268}	307
B1	$(2^4,3^9)$	$(1^7,2,3^8)$	{2: 1546, 3: 2247}	{1: -450, 2: -1836, 3: -2010}	391
B2	$(2^4,3^9)$	$(1^6,2^3,3^7)$	{2: 152, 3: 192}	{1: -51, 2: -148, 3: -150}	41
B1	$(2^4,3^5,4^4)$	$(1^7,2,3^4,4^4)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1836, 3: -2010, 4: -2268}	795
B1	$(2^3,3^{10})$	$(1^7,2,3^7,4)$	{2: 1546, 3: 2247}	{1: -450, 2: -1836, 3: -2010, 4: -2268}	684
B1	$(2^3,3^{10})$	$(1^7,3^9)$	{2: 1546, 3: 2247}	{1: -450, 3: -2010}	768
B2	$(2^3,3^{10})$	$(1^6,2^2,3^8)$	{2: 152, 3: 192}	{1: -51, 2: -147, 3: -150}	66
B1	$(2^3,3^7,4^3)$	$(1^7,2,3^4,4^4)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1836, 3: -2010, 4: -2268}	987
B2	$(2^3,3^6,4^4)$	$(1^6,2^2,3^4,4^4)$	{2: 152, 3: 192, 4: 216}	{1: -51, 2: -147, 3: -150, 4: -164}	46
B1	$(2^3,3,4^9)$	$(1^6,2,3,4^8)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1836, 3: -2010, 4: -2268}	549
B1	$(2^3,4^{10})$	$(1^6,2,4^9)$	{2: 1546, 4: 2756}	{1: -450, 2: -1836, 4: -2268}	650
B1	$(2^2,3^{11})$	$(1^7,2,3^7,5)$	{2: 1546, 3: 2247}	{1: -450, 2: -1480, 3: -2010, 5: -2502}	1357
B1	$(2^2,3^{11})$	$(1^7,2,3^6,4^2)$	{2: 1546, 3: 2247}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	1333
B1	$(2^2,3^{11})$	$(1^7,3^8,4)$	{2: 1546, 3: 2247}	{1: -450, 3: -2010, 4: -2268}	1061
B1	$(2^2,3^{11})$	$(1^6,2^2,3^7,4)$	{2: 1546, 3: 2247}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	561

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	$(2^2, 3^{11})$	$(1^6, 2, 3^9)$	{2: 1546, 3: 2247}	{1: -450, 2: -1480, 3: -2010}	289
B2	$(2^2, 3^{11})$	$(1^5, 2^3, 3^8)$	{2: 152, 3: 192}	{1: -51, 2: -136, 3: -150}	28
B1	$(2^2, 3^7, 4^4)$	$(1^6, 2, 3^5, 4^4)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	693
B1	$(2^2, 3^6, 4^5)$	$(1^6, 2, 3^4, 4^5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	794
B1	$(2^2, 3^5, 4^6)$	$(1^6, 2, 3^3, 4^6)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	895
B1	$(2^2, 3^4, 4^7)$	$(1^6, 2, 3^2, 4^7)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	996
B1	$(2^2, 3^3, 4^8)$	$(1^6, 2, 3, 4^8)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	1097
B1	$(2^2, 3^2, 4^9)$	$(1^6, 2, 3, 4^7, 5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 3: -2010, 4: -2268, 5: -2502}	1222
B1	$(2^2, 3^2, 4^9)$	$(1^6, 2, 4^9)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 4: -2268}	1198
B1	$(2^2, 3^2, 4^9)$	$(1^6, 3^2, 4^8)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	926
B1	$(2^2, 3^2, 4^9)$	$(1^5, 2^2, 3, 4^8)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1480, 3: -2010, 4: -2268}	426
B1	$(2^2, 3, 4^{10})$	$(1^6, 3, 4^9)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1027
B1	$(2^2, 4^{11})$	$(1^6, 3, 4^8, 5)$	{2: 1546, 4: 2756}	{1: -450, 3: -1905, 4: -2268, 5: -2502}	1257
B1	$(2^2, 4^{11})$	$(1^6, 4^{10})$	{2: 1546, 4: 2756}	{1: -450, 4: -2268}	1128
B1	$(2^2, 4^{11})$	$(1^5, 2, 3, 4^9)$	{2: 1546, 4: 2756}	{1: -450, 2: -1480, 3: -1905, 4: -2268}	461
B1	$(2, 3^{12})$	$(1^7, 3^8, 5)$	{2: 1546, 3: 2247}	{1: -450, 3: -2010, 5: -2502}	1378
B1	$(2, 3^{12})$	$(1^7, 3^7, 4^2)$	{2: 1546, 3: 2247}	{1: -450, 3: -2010, 4: -2268}	1354
B1	$(2, 3^{12})$	$(1^6, 2, 3^8, 4)$	{2: 1546, 3: 2247}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	741
B1	$(2, 3^{12})$	$(1^6, 3^{10})$	{2: 1546, 3: 2247}	{1: -450, 3: -2010}	310
B1	$(2, 3^{12})$	$(1^5, 2^2, 3^9)$	{2: 1546, 3: 2247}	{1: -450, 2: -1321, 3: -2010}	128
B1	$(2, 3^9, 4^3)$	$(1^6, 2, 3^5, 4^4)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	1044
B1	$(2, 3^8, 4^4)$	$(1^6, 2, 3^5, 4^3, 5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268, 5: -2502}	1169
B1	$(2, 3^8, 4^4)$	$(1^6, 2, 3^4, 4^5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	1145
B1	$(2, 3^8, 4^4)$	$(1^6, 3^6, 4^4)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	714

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	$(2,3^8,4^4)$	$(1^5,2^2,3^5,4^4)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	532
B1	$(2,3^7,4^5)$	$(1^6,2,3^4,4^4,5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268, 5: -2502}	1270
B1	$(2,3^7,4^5)$	$(1^6,2,3^3,4^6)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	1246
B1	$(2,3^7,4^5)$	$(1^6,3^5,4^5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	815
B1	$(2,3^7,4^5)$	$(1^5,2^2,3^4,4^5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	633
B1	$(2,3^6,4^6)$	$(1^6,2,3^3,4^5,5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268, 5: -2502}	1371
B1	$(2,3^6,4^6)$	$(1^6,2,3^2,4^7)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	1347
B1	$(2,3^6,4^6)$	$(1^6,3^4,4^6)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	916
B1	$(2,3^6,4^6)$	$(1^5,2^2,3^3,4^6)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	734
B1	$(2,3^5,4^7)$	$(1^6,3^3,4^7)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1017
B1	$(2,3^4,4^8)$	$(1^6,3^2,4^8)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1118
B1	$(2,3^3,4^9)$	$(1^6,3^2,4^7,5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1243
B1	$(2,3^3,4^9)$	$(1^6,3,4^9)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1219
B1	$(2,3^3,4^9)$	$(1^5,2,3^2,4^8)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	606
B1	$(2,3^2,4^{10})$	$(1^6,3,4^8,5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1344
B1	$(2,3^2,4^{10})$	$(1^6,4^{10})$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 4: -2268}	1320
B1	$(2,3^2,4^{10})$	$(1^5,2,3,4^9)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -2010, 4: -2268}	707
B1	$(2,3^2,4^5,5^5)$	$(1^5,2,3,4^4,5^5)$	{2: 1546, 3: 2247, 4: 2756, 5: 3195}	{1: -450, 2: -1321, 3: -2010, 4: -2268, 5: -2502}	982
B1	$(2,3,4^{11})$	$(1^5,2,3,4^8,5)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -1905, 4: -2268, 5: -2502}	937
B1	$(2,3,4^{11})$	$(1^5,2,4^{10})$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 4: -2268}	808
B1	$(2,3,4^{11})$	$(1^5,3^2,4^9)$	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 3: -1905, 4: -2268}	587

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	(2,3,4 ¹¹)	(1 ⁴ ,2 ² ,3,4 ⁹)	{2: 1546, 3: 2247, 4: 2756}	{1: -450, 2: -1321, 3: -1905, 4: -2268}	300
B1	(2,4 ¹²)	(1 ⁵ ,2,4 ⁹ ,5)	{2: 1546, 4: 2756}	{1: -450, 2: -1321, 4: -2268, 5: -2502}	933
B1	(2,4 ¹²)	(1 ⁵ ,3,4 ¹⁰)	{2: 1546, 4: 2756}	{1: -450, 3: -1584, 4: -2268}	904
B1	(2,4 ¹²)	(1 ⁴ ,2 ² ,4 ¹⁰)	{2: 1546, 4: 2756}	{1: -450, 2: -1321, 4: -2268}	296
B1	(2,4 ⁷ ,5 ⁵)	(1 ⁵ ,3,4 ⁵ ,5 ⁵)	{2: 1546, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1179
B1	(2,4 ⁶ ,5 ⁶)	(1 ⁵ ,3,4 ⁴ ,5 ⁶)	{2: 1546, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1234
B1	(2,4 ⁵ ,5 ⁷)	(1 ⁵ ,3,4 ³ ,5 ⁷)	{2: 1546, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1289
C1	(2,5 ¹²)	(1 ⁴ ,2,4,5 ¹⁰)	{2: 112, 5: 415}	{1: 0, 2: -138, 4: -232, 5: -322}	2
B1	(3 ¹³)	(1 ⁷ ,3 ⁸ ,6)	{3: 2247}	{1: -450, 3: -2010, 6: -2775}	1656
B1	(3 ¹³)	(1 ⁷ ,3 ⁷ ,4,5)	{3: 2247}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1671
B1	(3 ¹³)	(1 ⁷ ,3 ⁶ ,4 ³)	{3: 2247}	{1: -450, 3: -2010, 4: -2268}	1647
B1	(3 ¹³)	(1 ⁶ ,2,3 ⁸ ,5)	{3: 2247}	{1: -450, 2: -644, 3: -2010, 5: -2502}	1735
B1	(3 ¹³)	(1 ⁶ ,2,3 ⁷ ,4 ²)	{3: 2247}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1711
B1	(3 ¹³)	(1 ⁶ ,3 ⁹ ,4)	{3: 2247}	{1: -450, 3: -2010, 4: -2268}	603
B1	(3 ¹³)	(1 ⁵ ,2 ² ,3 ⁸ ,4)	{3: 2247}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1775
B1	(3 ¹³)	(1 ⁵ ,2,3 ¹⁰)	{3: 2247}	{1: -450, 2: -644, 3: -2010}	667
B1	(3 ¹³)	(1 ⁴ ,2 ³ ,3 ⁹)	{3: 2247}	{1: -450, 2: -644, 3: -2010}	1839
B1	(3 ¹¹ ,4 ²)	(1 ⁶ ,2,3 ⁵ ,4 ⁴)	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1913
B1	(3 ¹⁰ ,4 ³)	(1 ⁶ ,2,3 ⁵ ,4 ³ ,5)	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268, 5: -2502}	2038
B1	(3 ¹⁰ ,4 ³)	(1 ⁶ ,2,3 ⁴ ,4 ⁵)	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	2014
B1	(3 ¹⁰ ,4 ³)	(1 ⁶ ,3 ⁶ ,4 ⁴)	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	906
B1	(3 ¹⁰ ,4 ³)	(1 ⁵ ,2 ² ,3 ⁵ ,4 ⁴)	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	2078
B1	(3 ⁹ ,4 ⁴)	(1 ⁶ ,3 ⁶ ,4 ³ ,5)	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1031

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	$(3^9, 4^4)$	$(1^6, 3^5, 4^5)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1007
B1	$(3^9, 4^4)$	$(1^5, 2, 3^6, 4^4)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1071
B1	$(3^8, 4^5)$	$(1^6, 3^5, 4^4, 5)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1132
B1	$(3^8, 4^5)$	$(1^6, 3^4, 4^4, 6)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1108
B1	$(3^8, 4^5)$	$(1^5, 2, 3^5, 4^5)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1172
B1	$(3^7, 4^6)$	$(1^6, 3^4, 4^4, 5, 5)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1233
B1	$(3^7, 4^6)$	$(1^6, 3^3, 4^4, 7)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1209
B1	$(3^7, 4^6)$	$(1^5, 2, 3^4, 4^4, 6)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1273
B1	$(3^6, 4^7)$	$(1^6, 3^3, 4^4, 6, 5)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1334
B1	$(3^6, 4^7)$	$(1^6, 3^2, 4^4, 8)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1310
B1	$(3^6, 4^7)$	$(1^5, 2, 3^3, 4^4, 7)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1374
B1	$(3^5, 4^8)$	$(1^6, 3^2, 4^4, 7, 5)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1435
B1	$(3^5, 4^8)$	$(1^6, 3, 4^4, 9)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	1411
B1	$(3^5, 4^8)$	$(1^5, 2, 3^2, 4^4, 8)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1475
B1	$(3^4, 4^9)$	$(1^6, 3^2, 4^4, 7, 6)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 6: -2775}	1521
B1	$(3^4, 4^9)$	$(1^6, 3^2, 4^4, 6, 5^2)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1560
B1	$(3^4, 4^9)$	$(1^6, 3, 4^4, 8, 5)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	1536
B1	$(3^4, 4^9)$	$(1^6, 4^4, 10)$	{3: 2247, 4: 2756}	{1: -450, 4: -2268}	1512
B1	$(3^4, 4^9)$	$(1^5, 2, 3^2, 4^4, 7, 5)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268, 5: -2502}	1600
B1	$(3^4, 4^9)$	$(1^5, 2, 3, 4^4, 9)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1576
B1	$(3^4, 4^9)$	$(1^5, 3^3, 4^4, 8)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	468
B1	$(3^4, 4^9)$	$(1^4, 2^2, 3^2, 4^4, 8)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1640
B1	$(3^3, 4^4, 10)$	$(1^5, 2, 3, 4^4, 8, 5)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268, 5: -2502}	1701

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	$(3^3, 4^{10})$	$(1^5, 2, 4^{10})$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 4: -2268}	1677
B1	$(3^3, 4^{10})$	$(1^5, 3^2, 4^9)$	{3: 2247, 4: 2756}	{1: -450, 3: -2010, 4: -2268}	569
B1	$(3^3, 4^{10})$	$(1^4, 2^2, 3, 4^9)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -2010, 4: -2268}	1741
B1	$(3^3, 4^5, 5^5)$	$(1^5, 3^2, 4^4, 5^5)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	844
B1	$(3^3, 4^4, 5^6)$	$(1^5, 3^2, 4^3, 5^6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -2010, 4: -2268, 5: -2502}	899
B1	$(3^2, 4^{11})$	$(1^5, 2, 3, 4^8, 6)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -1905, 4: -2268, 6: -2775}	1892
B1	$(3^2, 4^{11})$	$(1^5, 2, 3, 4^7, 5^2)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -1905, 4: -2268, 5: -2502}	1931
B1	$(3^2, 4^{11})$	$(1^5, 2, 4^9, 5)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 4: -2268, 5: -2502}	1802
B1	$(3^2, 4^{11})$	$(1^5, 3^2, 4^8, 5)$	{3: 2247, 4: 2756}	{1: -450, 3: -1905, 4: -2268, 5: -2502}	904
B1	$(3^2, 4^{11})$	$(1^5, 3, 4^{10})$	{3: 2247, 4: 2756}	{1: -450, 3: -1905, 4: -2268}	775
B1	$(3^2, 4^{11})$	$(1^4, 2^2, 3, 4^8, 5)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -1905, 4: -2268, 5: -2502}	1971
B1	$(3^2, 4^{11})$	$(1^4, 2^2, 4^{10})$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 4: -2268}	1842
B1	$(3^2, 4^{11})$	$(1^4, 2, 3^2, 4^9)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -1905, 4: -2268}	944
B1	$(3^2, 4^{11})$	$(1^3, 2^3, 3, 4^9)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -1905, 4: -2268}	2011
B1	$(3^2, 4^6, 5^5)$	$(1^5, 3, 4^5, 5^5)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1905, 4: -2268, 5: -2502}	1050
B1	$(3^2, 4^5, 5^6)$	$(1^5, 3, 4^4, 5^6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1905, 4: -2268, 5: -2502}	1105
B1	$(3^2, 4^4, 5^7)$	$(1^5, 3, 4^3, 5^7)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1905, 4: -2268, 5: -2502}	1160
B1	$(3^2, 4^3, 5^8)$	$(1^5, 3, 4^2, 5^8)$	{3: 2247, 4: 2756, 5: 3114}	{1: -450, 3: -1905, 4: -2268, 5: -2502}	567
B1	$(3^2, 4, 5^{10})$	$(1^4, 2, 3, 4, 5^9)$	{3: 2247, 4: 2756, 5: 3114}	{1: -450, 2: -644, 3: -1905, 4: -2268, 5: -2502}	555
B1	$(3^2, 5^{11})$	$(1^4, 2, 3, 5^{10})$	{3: 2247, 5: 3114}	{1: -450, 2: -644, 3: -1905, 5: -2502}	529
B1	$(3, 4^{12})$	$(1^5, 2, 4^9, 6)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 4: -2268, 6: -2775}	1888

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	$(3,4^{12})$	$(1^5, 2, 4^8, 5^2)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 4: -2268, 5: -2502}	1927
B1	$(3,4^{12})$	$(1^5, 3, 4^9, 5)$	{3: 2247, 4: 2756}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1221
B1	$(3,4^{12})$	$(1^5, 4^{11})$	{3: 2247, 4: 2756}	{1: -450, 4: -2268}	771
B1	$(3,4^{12})$	$(1^4, 2^2, 4^9, 5)$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 4: -2268, 5: -2502}	1967
B1	$(3,4^{12})$	$(1^4, 2, 3, 4^{10})$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 3: -1584, 4: -2268}	1261
B1	$(3,4^{12})$	$(1^3, 2^3, 4^{10})$	{3: 2247, 4: 2756}	{1: -450, 2: -644, 4: -2268}	2007
B1	$(3,4^8, 5^4)$	$(1^5, 3, 4^5, 5^5)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1441
B1	$(3,4^7, 5^5)$	$(1^5, 3, 4^5, 5^4, 6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502, 6: -2775}	1457
B1	$(3,4^7, 5^5)$	$(1^5, 3, 4^4, 5^6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1496
B1	$(3,4^7, 5^5)$	$(1^5, 4^6, 5^5)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502}	1046
B1	$(3,4^7, 5^5)$	$(1^4, 2, 3, 4^5, 5^5)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 2: -644, 3: -1584, 4: -2268, 5: -2502}	1536
B1	$(3,4^6, 5^6)$	$(1^5, 3, 4^4, 5^5, 6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502, 6: -2775}	1512
B1	$(3,4^6, 5^6)$	$(1^5, 3, 4^3, 5^7)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1551
B1	$(3,4^6, 5^6)$	$(1^5, 4^5, 5^6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502}	1101
B1	$(3,4^6, 5^6)$	$(1^4, 2, 3, 4^4, 5^6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 2: -644, 3: -1584, 4: -2268, 5: -2502}	1591
B1	$(3,4^5, 5^7)$	$(1^5, 3, 4^3, 5^6, 6)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 3: -1584, 4: -2268, 5: -2502, 6: -2775}	1567
B1	$(3,4^5, 5^7)$	$(1^5, 3, 4^2, 5^8)$	{3: 2247, 4: 2756, 5: 3114}	{1: -450, 3: -1584, 4: -2268, 5: -2502}	1039
B1	$(3,4^5, 5^7)$	$(1^5, 4^4, 5^7)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502}	1156
B1	$(3,4^5, 5^7)$	$(1^4, 2, 3, 4^3, 5^7)$	{3: 2247, 4: 2756, 5: 3195}	{1: -450, 2: -644, 3: -1584, 4: -2268, 5: -2502}	1646
B1	$(3,4^2, 5^{10})$	$(1^4, 2, 4^2, 5^9)$	{3: 2247, 4: 2756, 5: 3114}	{1: -450, 2: -644, 4: -2268, 5: -2502}	551
B1	$(3,5^{12})$	$(1^4, 3, 4, 5^{10})$	{3: 2247, 5: 3114}	{1: -450, 3: -1584, 4: -1944, 5: -2502}	117

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	$(4^{\wedge}13)$	$(1^{\wedge}5,3,4^{\wedge}9,6)$	{4: 2756}	{1: -450, 3: -1340, 4: -2268, 6: -2775}	1551
B1	$(4^{\wedge}13)$	$(1^{\wedge}5,3,4^{\wedge}8,5^{\wedge}2)$	{4: 2756}	{1: -450, 3: -1340, 4: -2268, 5: -2502}	1590
B1	$(4^{\wedge}13)$	$(1^{\wedge}5,4^{\wedge}10,5)$	{4: 2756}	{1: -450, 4: -2268, 5: -2502}	896
B1	$(4^{\wedge}13)$	$(1^{\wedge}4,2,3,4^{\wedge}9,5)$	{4: 2756}	{1: -450, 2: -644, 3: -1340, 4: -2268, 5: -2502}	1630
B1	$(4^{\wedge}13)$	$(1^{\wedge}4,2,4^{\wedge}11)$	{4: 2756}	{1: -450, 2: -644, 4: -2268}	936
B1	$(4^{\wedge}13)$	$(1^{\wedge}4,3^{\wedge}2,4^{\wedge}10)$	{4: 2756}	{1: -450, 3: -1340, 4: -2268}	1168
B1	$(4^{\wedge}13)$	$(1^{\wedge}3,2^{\wedge}2,3,4^{\wedge}10)$	{4: 2756}	{1: -450, 2: -644, 3: -1340, 4: -2268}	1670
B1	$(4^{\wedge}10,5^{\wedge}3)$	$(1^{\wedge}5,3,4^{\wedge}5,5^{\wedge}5)$	{4: 2756, 5: 3195}	{1: -450, 3: -1340, 4: -2268, 5: -2502}	1755
B1	$(4^{\wedge}9,5^{\wedge}4)$	$(1^{\wedge}5,4^{\wedge}6,5^{\wedge}5)$	{4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502}	1116
B1	$(4^{\wedge}8,5^{\wedge}5)$	$(1^{\wedge}5,4^{\wedge}6,5^{\wedge}4,6)$	{4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502, 6: -2775}	1132
B1	$(4^{\wedge}8,5^{\wedge}5)$	$(1^{\wedge}5,4^{\wedge}5,5^{\wedge}6)$	{4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502}	1171
B1	$(4^{\wedge}8,5^{\wedge}5)$	$(1^{\wedge}4,2,4^{\wedge}6,5^{\wedge}5)$	{4: 2756, 5: 3195}	{1: -450, 2: -644, 4: -2268, 5: -2502}	1211
B1	$(4^{\wedge}7,5^{\wedge}6)$	$(1^{\wedge}5,4^{\wedge}5,5^{\wedge}5,6)$	{4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502, 6: -2775}	1187
B1	$(4^{\wedge}7,5^{\wedge}6)$	$(1^{\wedge}5,4^{\wedge}4,5^{\wedge}7)$	{4: 2756, 5: 3195}	{1: -450, 4: -2268, 5: -2502}	1226
B1	$(4^{\wedge}7,5^{\wedge}6)$	$(1^{\wedge}4,2,4^{\wedge}5,5^{\wedge}6)$	{4: 2756, 5: 3195}	{1: -450, 2: -644, 4: -2268, 5: -2502}	1266
B1	$(4^{\wedge}6,5^{\wedge}7)$	$(1^{\wedge}4,2,4^{\wedge}4,5^{\wedge}7)$	{4: 2756, 5: 3195}	{1: -450, 2: -644, 4: -2268, 5: -2502}	1321
B1	$(4^{\wedge}5,5^{\wedge}8)$	$(1^{\wedge}4,2,4^{\wedge}3,5^{\wedge}8)$	{4: 2756, 5: 3114}	{1: -450, 2: -644, 4: -2268, 5: -2502}	728
B1	$(4^{\wedge}4,5^{\wedge}9)$	$(1^{\wedge}4,2,4^{\wedge}2,5^{\wedge}9)$	{4: 2756, 5: 3114}	{1: -450, 2: -644, 4: -2268, 5: -2502}	702
B1	$(4^{\wedge}3,5^{\wedge}10)$	$(1^{\wedge}4,3,4^{\wedge}2,5^{\wedge}9)$	{4: 2756, 5: 3114}	{1: -450, 3: -1340, 4: -2268, 5: -2502}	214
B1	$(4^{\wedge}2,5^{\wedge}11)$	$(1^{\wedge}4,3,4,5^{\wedge}10)$	{4: 2756, 5: 3114}	{1: -450, 3: -1340, 4: -2154, 5: -2502}	302
B1	$(4,5^{\wedge}12)$	$(1^{\wedge}4,3,4,5^{\wedge}9,6)$	{4: 2756, 5: 3114}	{1: -450, 3: -1340, 4: -1944, 5: -2502, 6: -2775}	447
B1	$(4,5^{\wedge}12)$	$(1^{\wedge}4,3,5^{\wedge}11)$	{4: 2756, 5: 3114}	{1: -450, 3: -1340, 5: -2502}	162

Template	Demand s	Residual rho	Label minima ell_s	Source minima sigma_rho	Positive gap
B1	(4,5 ¹²)	(1 ⁴ ,4 ² ,5 ¹⁰)	{4: 2756, 5: 3114}	{1: -450, 4: -1944, 5: -2502}	116
B1	(4,5 ¹²)	(1 ³ ,2,3,4,5 ¹⁰)	{4: 2756, 5: 3114}	{1: -450, 2: -644, 3: -1340, 4: -1944, 5: -2502}	526
B1	(5 ¹³)	(1 ⁴ ,3,5 ¹⁰ ,6)	{5: 3114}	{1: -450, 3: -1340, 5: -2502, 6: -2775}	97
B1	(5 ¹³)	(1 ⁴ ,4,5 ¹¹)	{5: 3114}	{1: -450, 4: -1608, 5: -2502}	102
B1	(5 ¹³)	(1 ³ ,2,3,5 ¹¹)	{5: 3114}	{1: -450, 2: -644, 3: -1340, 5: -2502}	176

Review boundary

The candidate theorem follows from the implications and finite arithmetic printed above. The companion records where these texts originated, which historical dependencies were removed, and what the internal checks established. Independent specialist review of the bridge, both clipping arguments, completeness of the tables and every endpoint inequality remains OPEN.